

Leveraging Polymeric Design for Marine Anti-Biofouling: Mechanism, System, and Future

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Marine biofouling presents major challenges for marine industries, leading to substantial economic losses and increased energy consumption. Historically, antifouling strategies relied on coatings containing tributyltin (TBT) or metal- and organic-based biocides. While effective, these coatings pose considerable environmental and biological risks, highlighting the need for sustainable and non-toxic alternatives. Recent advances in polymer science have opened new avenues for environmentally friendly antifouling solutions. Various approaches such as dynamic coatings (chemical and physical), fouling-release coatings, fouling-resistant coatings, and other emerging strategies leveraging polymeric design to prevent organism attachment have been developed. Yet, comprehensive and state-of-the-art overviews focusing specifically on their design from a polymer science perspective remain limited. This review first summarizes recent progress in marine antifouling polymeric materials and interfacial designs, focusing on the relationship between polymer structure and functional performance. It then discusses the current challenges associated with these materials and proposes potential solutions for further development. Finally, the review outlines possible directions for future progress in the field, aiming to provide readers with a comprehensive understanding of marine antifouling polymeric materials and interfaces while inspiring continued research and innovation in sustainable, durable, and non-toxic antifouling coatings.

mechanical cleaning, seawater electrolysis, ultraviolet (UV) irradiation, and ultrasonic treatments.^[6–8,9] Among these, antifouling coatings remain the most practical, cost-effective, and widely adopted solution.^[10,11]

In the 20th century, tributyltin (TBT)-based antifouling coatings were widely applied in commercial and recreational shipping owing to their long-lasting effectiveness. However, owing to their severe environmental and ecological impacts, TBT coatings were globally banned in 2008.^[12] Since then, metal biocides such as cuprous oxide, along with booster biocides such as copper pyrithione, zineb, and diuron, have been used in marine antifouling coatings.^[13,14] Although these biocide-based coatings provide effective protection against biofouling, their toxicity and long-term use continue to pose risks to marine environments and ecosystems.^[15,16] Therefore, with increasing societal concern for the marine environment and ecosystem, the future of biocide-containing coatings remains uncertain because they may ultimately face the same restrictions as TBT.^[13] This pressing need to combat marine biofouling effectively without releasing

toxic substances into aquatic systems has spurred the development of durable, sustainable, and non-toxic antifouling polymeric materials and interfaces.^[17]

With advances in polymer science, novel designs of polymeric materials and interfaces have opened new opportunities for developing effective marine antifouling coatings. In recent years, various environmentally friendly antifouling coatings and strategies have been introduced, including dynamic coatings (chemical and physical), fouling-release coatings (FRCs), fouling-resistant coatings, and other emerging approaches (**Figure 1**). Specifically, our group has made considerable contributions in this field, particularly in the areas of dynamic coatings, FRCs, and fouling-resistant coatings.^[10,18–20,21] However, although several reviews have discussed the mechanisms of marine biofouling formation, antifouling technologies, and bioinspired strategies.^[22,23] Comprehensive overviews focusing specifically on marine antifouling coatings design from a polymer science perspective—and on the relationship between polymer structure and antifouling performance—remain limited.^[24,25] This review aims to fill this gap by specifically focusing on leveraging polymeric design for marine anti-biofouling, which extends beyond the existing

1. Introduction

The accumulation of marine biofouling—comprising microorganisms, animals, and plants—on surfaces poses serious challenges to marine industries and infrastructures.^[1,2] For example, biofouling on ship hulls increases drag during navigation, resulting in greater fuel consumption.^[3] In heat exchangers, fouling accumulation clogs pipelines and drastically reduces heat transfer efficiency.^[4] In aquaculture, fouling organisms attached to cage netting impede nutrient exchange, thereby affecting both productivity and sustainability.^[5] Over the years, various strategies have been developed to mitigate marine fouling, including coatings,

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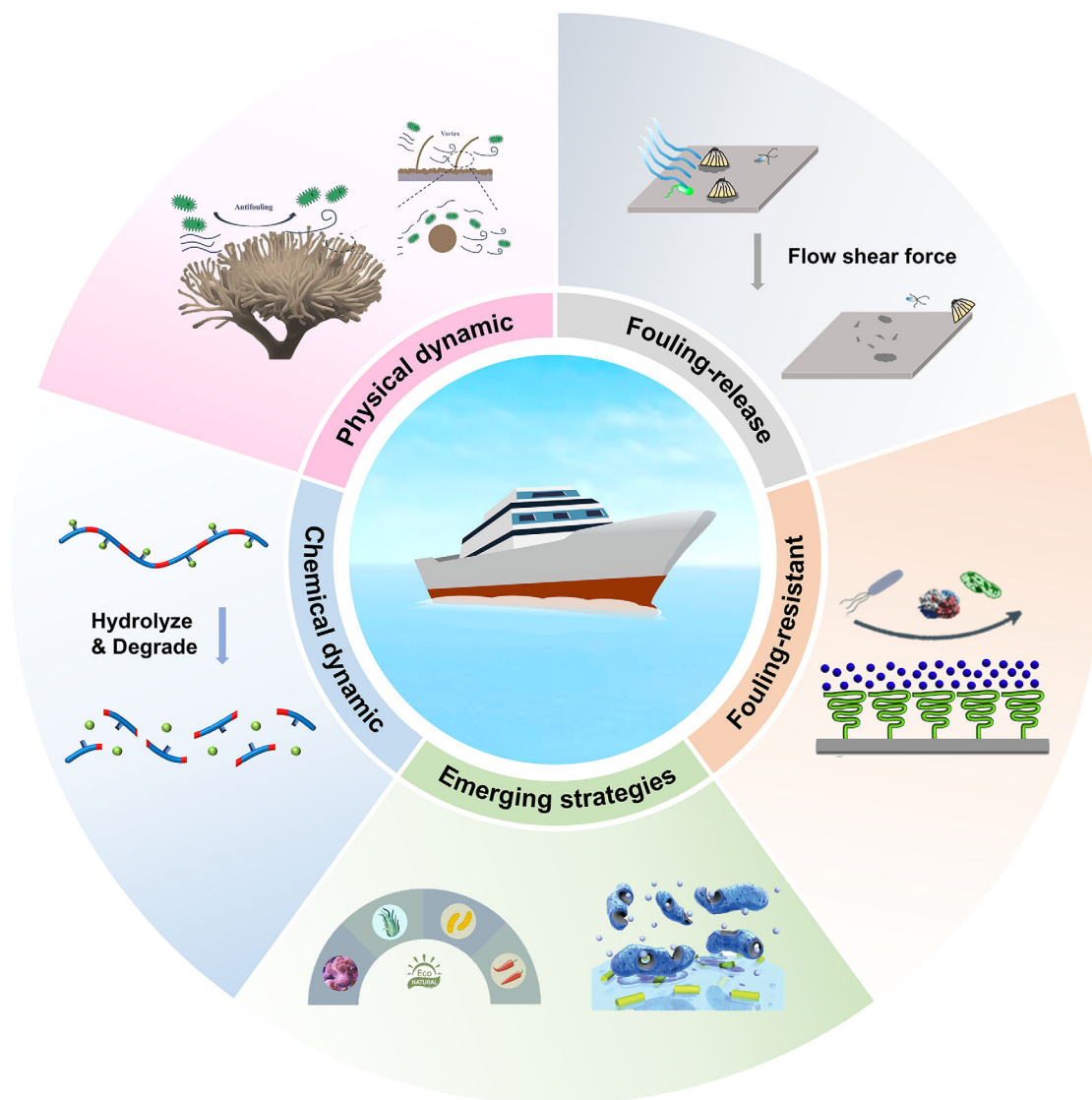


Figure 1. Overview of non-toxic marine antifouling coatings and strategies. [The figures used here have been adapted with permission from the original sources cited in this review].

literature in terms of conceptual depth and coverage. Furthermore, with the expansion of marine industries, driven by increasing shipping activities, offshore infrastructure, and renewable energy systems, the demand for durable and highly effective antifouling coatings has risen considerably.^[26] Finally, in line with the principles of sustainable development, there is a growing emphasis on the development of eco-friendly antifouling materials and strategies. The urgency to minimize toxic substances in antifouling coatings has never been greater, driven by stricter regulations and increased public awareness.^[17] Therefore, the convergence of advances in polymer science, growing demands from the marine industry, and environmental protection requirements underscores the need for a comprehensive, state-of-the-art review on polymeric materials and interfaces for marine anti-biofouling.

This review provides an overview of recent advances in marine antifouling polymeric materials and interfaces, highlighting

the relationship between their structures and properties. Additionally, it addresses future challenges and opportunities for developing sustainable and non-toxic antifouling coatings, offering potential solutions for future research and innovation. Finally, the review outlines prospective directions for further advancements in the field, aiming to give readers a comprehensive understanding of marine antifouling polymeric materials and interfaces while inspiring continued development and innovation.

2. Dynamic Surface Antifouling

Since TBT self-polishing copolymer (SPC) coatings were banned by the International Maritime Organization in 2008, tin-free SPC coatings containing specialized acrylate copolymers—such as silyl acrylate, copper acrylate, and zinc acrylate—have been developed as alternative strategies. The side groups in these

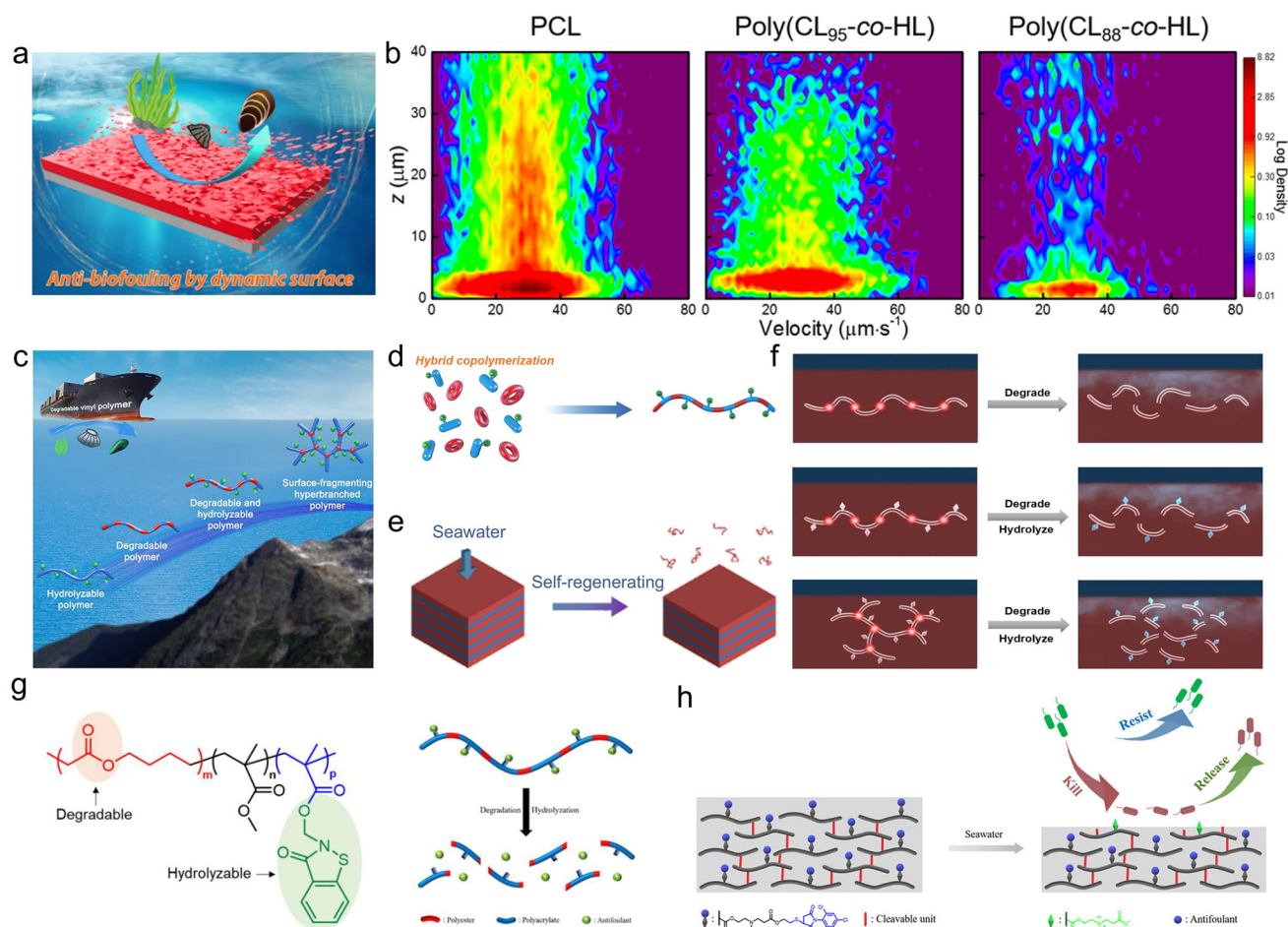


Figure 2. a) Schematic of anti-biofouling by a dynamic surface. Reproduced with permission.^[18] Copyright 2019, The Royal Society of Chemistry. b) Density distribution (heat maps) of instantaneous 3D velocities and distances from the surface (z) for *E. coli* on PCL, poly(CL₉₅-co-HL), and poly(CL₈₈-co-HL) surfaces. Color bars represent the relative density of data points on a logarithmic scale. Reproduced with permission.^[31] Copyright 2017, American Chemical Society. c) Structural schematic of hydrolyzable polymer, degradable polymer, degradable and hydrolyzable polymer, and surface-fracturing hyperbranched polymer. Reproduced with permission.^[10] Copyright 2022, American Chemical Society. d) Schematic of hybrid copolymerization. Reproduced with permission.^[10] Copyright 2022, American Chemical Society. e) Schematic of self-regenerating mechanisms of chemical dynamic coatings. f) Anti-biofouling mechanisms of degradable polymer, degradable and hydrolyzable polymer, and surface-fracturing hyperbranched polymer. Reproduced with permission.^[10] Copyright 2022, American Chemical Society. g) Structure of poly(ester-co-acrylate) with BIT pendant groups and their degradation and hydrolyzation mechanism. Reproduced with permission.^[10] Copyright 2022, American Chemical Society. h) Hyperbranched polymer with kill-resist-renew trinity for antifouling coatings. Reproduced with permission.^[59] Copyright 2021, American Chemical Society.

SPCs, including silyl, copper, and zinc esters, undergo hydrolysis when exposed to seawater and are subsequently polished away by strong water flow. These SPC coatings demonstrate effective anti-biofouling performance, particularly when vessels navigate at high speeds. However, these copolymer-based coatings exhibit poor antifouling performance under static conditions. In addition, the main chain of SPC is nondegradable, leading to the generation of marine microplastics when polymer fragments enter the ocean.^[10,27] It has been estimated that 400–1194 tons of marine coatings are introduced into European waters annually, $\approx 40\%$ of which use nondegradable polymers as binders. According to the International Union for Conservation of Nature, marine coatings contribute $\approx 3.7\%$ of primary microplastic releases into the world's oceans.^[28–30] Thus, increasing societal concern over environmental impacts and biofouling has created a pressing need for the development of environmentally friendly and

effective anti-biofouling coatings. Dynamic surfaces, which can prevent the settlement and adhesion of marine fouling organisms through continuous surface changes without releasing any heavy metal-based substances, represent a promising approach for eco-friendly antifouling materials (Figure 2a). Typically, dynamic surfaces are classified into two categories: chemical dynamic coatings and physical dynamic coatings.

Chemical dynamic coatings maintain continuously changing surfaces through main-chain degradation, offering tunable renewability and excellent antifouling performance. To elucidate the antifouling mechanism of such dynamic coatings, Qi et al. studied bacterial movement trajectories on a dynamic surface using digital holographic microscopy.^[31] The dynamic surface was constructed from a biodegradable copolymer, poly(CL-co-HL), composed of ϵ -caprolactone (CL) and δ -hexalactone (HL), with the degradation rate increasing alongside HL content. This

biodegradable polymer sustains a dynamic surface in seawater through ongoing hydrolytic and enzymatic degradation. The results in Figure 2b clearly show that as the degradation rate increases, bacteria are more likely to escape from the surface, resulting in a reduced bacterial population. Therefore, a dynamic surface with an appropriate rate of change can effectively inhibit bacterial adhesion.^[20] This strategy is actually similar to the shedding behaviors of some natural species. In marine environments, certain species—such as crustose coralline algae and large animals such as pilot whales—can remove fouling organisms from their surfaces through epithelium shedding.^[32–35,36] Besides shedding defense mechanism, some organisms, including dolphins and corals, have soft skin that forms an unstable surface under marine flow. This continuously dynamic surface hinders bacterial attachment, and any fouling organisms that do adhere can be readily removed through surface deformation under flow.^[37] Similar to this strategy, various physical dynamic antifouling coatings have also been developed, whose surface and topography can continuously change in response to external stimuli such as turbulent flows, electrical voltage, or air pressure, facilitating the detachment of bacterial biofilms and macrofouling organisms.^[38] The following sections review research efforts and recent advances in dynamic surfaces for antifouling applications.

2.1. Chemical Dynamic Antifouling Coatings

Chemical dynamic antifouling coatings are designed to maintain a continuously self-renewing surface in seawater through the degradation of polyester-based polymers, thereby preventing biofouling adhesion. Unlike traditional SPCs with nondegradable hydrocarbon main chains, polyester main chains can degrade into small molecules, helping to reduce the accumulation of microplastics in the ocean.^[39–41] Studies have shown that higher polymer degradation rates correspond to improved antifouling performance.^[31] Based on this approach, a series of degradable polyester-based polymers has been developed (Figure 2c).^[18]

Traditional polyesters such as poly(ϵ -caprolactone) (PCL), poly(butylene succinate) (PBS), and poly(L-lactic acid) (PLA) cannot be directly used as coatings owing to poor film-forming ability and low adhesion strength. Additionally, their high crystallinity results in slow degradation, limiting their antifouling effectiveness.^[18,42,43] Combining urethane moieties with degradable polymers can enhance both film-forming properties and adhesion strength. For example, Ma et al. synthesized a PCL-based polyurethane, where PCL acts as the soft segment and lysine ethyl ester diisocyanate and 1,4-butanediol serve as the hard segments. The coating exhibited improved adhesion strength, yet its crystallinity is still high, resulting in a low degradation rate. To lower the crystallinity, they used a dihydroxyl-terminated polyester copolymer of caprolactone (CL) and glycolide (GA) as a soft segment for polyurethane, disrupting the regularity of polymer chains and thereby reducing crystallinity.^[18] Polyurethane based on the CL–GA copolymer exhibits improved antifouling performance owing to its accelerated degradation rate. Furthermore, 4,5-dichloro-2-*n*-octyl-4-isothiazolin-3-one (DCOIT) was incorporated into the degradable polymers, creating a system with enhanced antifouling performance and extended service life.^[39]

Chen et al. introduced various polyesters into polyurethane and found that adjusting the ester density allows regulation of degradability, while the resulting degradable polyurethane maintains excellent adhesion strength owing to hydrogen bonding between the substrate and urethane moieties.^[44] Yao et al. prepared a composite coating by blending clay with PCL; the clay reduced the spherulite size of PCL, increasing the amorphous regions and considerably accelerating both enzymatic and hydrolytic degradation compared to pure PCL. By further incorporating DCOIT into the composite, the coating was able to resist biofouling for up to ten months.^[41]

Similar to polyurethanes, polyacrylates are suitable for use as adhesives and coatings owing to their excellent film-forming properties on various substrates.^[18] For instance, Zhang et al. prepared hyperbranched PCL with glycidyl methacrylate and poly(CL-co-methacrylic acid), where the hydrophilic methacrylic acid accelerates ester bond degradation.^[45,46] Additionally, poly(ester-co-acrylate) copolymers synthesized via radical ring-opening polymerization (RROP) have been explored. Dai et al. prepared a copolymer of 2-methylene-1,3-dioxepane (MDO) and methyl methacrylate (MMA); this amorphous copolymer exhibits excellent film-forming properties and strong adhesion to substrates. When combined with DCOIT, the resulting coating demonstrates effective antifouling performance.^[47] Xie et al. also synthesized a copolymer of MDO, 2-(dimethylamino) ethyl methacrylate (DEM), and 3-(methacryloxypropyl) trimethoxysilane (KH570). This polymer exhibits antifouling properties owing to its degradability and the hydrophilic nature of DEM.^[48]

Notably, the antifouling performance of degradable polymers improves with increasing ester density and hydrophilicity but decreases with higher crystallinity, a trend that contrasts with their mechanical properties. Therefore, achieving simultaneous control over both degradation and mechanical performance in degradable polymers is limited. Moreover, traditional degradable polymers are difficult to functionalize or chemically modify because cyclic monomers generally lack reactive or modifiable groups.^[10] To overcome these challenges, functional vinyl polymers and degradable polymers have been integrated into a single polymer chain through hybrid copolymerization (Figure 2d). These polymer chains feature a cleavable main chain and hydrolyzable side groups, such as silyl esters, antifoulants, and hydrolysis-induced zwitterions (HIZs).^[49,50] As shown in Figure 2e,f, the hydrolyzable side groups convert from hydrophobic to hydrophilic upon hydrolysis, enhancing both the self-polishing rate and the antifouling performance of the coating. Moreover, the polymeric coatings can self-regenerate their surfaces under seawater exposure owing to ongoing degradation and hydrolysis. Compared to traditional degradable polymers, degradable vinyl polymers offer more controllable degradation rates and greater ease of functionalization.

Zhou et al. copolymerized MMA, tributyl silyl methacrylate (TBSM), and MDO to produce a polymer featuring a degradable backbone and hydrolyzable side groups. The degradation and hydrolysis rates could be tuned by adjusting the molar ratio of ester units to tributyl silyl ester side groups.^[51] Additionally, to investigate the effect of leaving-group structure on hydrolysis rates, Xie et al. copolymerized MDO with bis(trimethylsiloxy)methylsilyl methacrylate (MATM2), triisopropylsilyl methacrylate (TIPSM), and TBSM, respectively. They found that as steric hindrance

decreased, the hydrolysis rate of the silyl ester increased.^[52] To further extend the coating's service life and enhance its antifouling performance, environmentally friendly antifoulants can be chemically incorporated. For instance, Dai et al. developed a novel degradable copolymer with antifoulant pendant groups via RROP of MMA, MDO, and *N*-methacryloyloxy methyl benzothiazolinone (V-BIT) (Figure 2g).^[49] The eco-friendly BIT was chosen as an antifoulant owing to its high efficiency, broad-spectrum antimicrobial activity, and biodegradability.^[53,54] The polymer undergoes side-group hydrolysis and main-chain degradation, generating a continuously renewed surface while releasing antifoulants. Therefore, antifoulant release can be conveniently tuned by controlling both the main-chain degradation and side-group hydrolysis. The copolymer demonstrates both tunable enzymatic and hydrolytic degradation and high antifouling efficacy against marine biofouling, resulting from the synergistic effect of surface renewal and antifoulant release. The side groups of vinyl polymers can be functionalized not only with antifoulants but also with HIZs, which have been shown to inhibit the adhesion of marine bacteria, algae, and diatoms.^[55] By combining the self-renewal capability of silyl acrylate polymers with the protein resistance of zwitterionic polymers, a novel self-generating and self-renewing zwitterionic coating for marine anti-biofouling has been developed.^[50] The coating was prepared by copolymerizing a hydrophobic hydrolysis-induced zwitterionic monomer—tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA)—with MMA and MDO. Upon hydrolysis, the coating can self-generate zwitterions on its surface, providing persistent fouling resistance. The ester units in the main chain render the polymers degradable, enabling continuous surface self-renewal. This self-generating and self-renewing zwitterionic coating exhibits excellent protein resistance and antifouling performance against marine biofouling, making it a promising candidate for marine industry applications.

To further enhance the dynamic and antifouling performance of degradable coatings under static conditions, hyperbranched polymers have been introduced. These polymers provide more degradable sites, producing lower molecular weight fragments upon degradation compared to linear degradable polymers. Additionally, hyperbranched polymers facilitate surface self-renewal while maintaining high solubility—allowing the preparation of coatings with high solid content but low viscosity—and excellent film-forming properties.^[2,10,45,56] For example, Mei et al. synthesized a hyperbranched copolymer via reversible addition–fragmentation chain transfer polymerization of divinyl-functionalized poly(ϵ -caprolactone) (PCL- V_2), tertiary carboxybetaine ester (TCB), and MMA.^[57] Upon hydrolysis of TCB, zwitterions are generated to resist biofouling adhesion, while the polymer's branching structure reduces crystallinity and enhances the degradation of PCL segments, enabling surface renewal. This polymer features a fragmenting surface that allows continuous self-regeneration of zwitterions, providing both fouling resistance and renewability. Moreover, the polymer's degradation rate can be finely tuned by adjusting the PCL content. This surface-fragmenting, self-regenerating hyperbranched zwitterionic copolymer offers versatile applications in antifouling coatings.

Zwitterions in the coating prevent biofouling adhesion, representing a passive antifouling strategy.^[58] To further enhance

the antifouling performance of hyperbranched polymer coatings, an attack-and-defense strategy was developed by copolymerizing a novel monomer—tertiary carboxybetaine ester acrylate—with the antifouling group *N*-(2,4,6-trichlorophenyl)maleimide (TCB–TCPM) and methacrylic anhydride.^[59] Upon hydrolysis, TCB–TCPM on the coating surface releases TCPM to actively kill biofouling (attack), while simultaneously generating zwitterionic groups to resist biofouling adhesion (defense) (Figure 2h). As a result, the coating can transition from “attacking” to “defending” on the surface while maintaining strong matrix integrity. Additionally, the surface of the coating degrades into small fragments in seawater, creating a dynamic surface that continuously regenerates the antifouling layer. The hyperbranched dynamic coating features a controllable degradation rate, which increases with the degree of branching, and demonstrates excellent antifouling performance through the synergistic effects of “attack” and “defense.”

In summary, chemical dynamic coatings exhibit excellent antifouling performance owing to main-chain degradation, which produces a continuously self-renewing surface. By combining main-chain degradation with side-chain hydrolysis, the polymer degradation rate can be controlled, further enhancing antifouling effectiveness. Notably, introducing a hyperbranched structure provides additional degradable sites, generating small molecules that disperse easily in seawater and do not contribute to marine microplastic formation. These coatings also maintain outstanding antifouling performance in stagnant water. As discussed above, these chemical dynamic coatings mainly consist of degradable polyesters, whose ester linkages are susceptible to cleave in the presence of water and enzyme, generating lower molecular weight products such as acids and alcohols.^[60] Once taken up by marine microbes, they can be absorbed and utilized as carbon and energy sources, ultimately being broken down into simple inorganic compounds, such as carbon dioxide, water, and biomass.^[60,61] Actually, the potential ecotoxicological consequences of degradation products have been assessed using algae growth inhibition assays. The average optical density (OD_{680}) values of the *Chlorella* solution in the experimental groups showed a consistent trend with those of the blank control groups, confirming the biocompatibility of degradation products with marine organisms.^[62] However, despite these obvious advantages, chemical dynamic coatings are consumable, and their service life is limited by coating thickness. To achieve superior antifouling performance, the coating must degrade rapidly enough, which inherently shortens its lifespan. Integrating degradable polymers with environmentally friendly antifoulants offers a potential strategy to extend service life while maintaining effectiveness.

2.2. Physical Dynamic Antifouling Coatings

Unlike chemical degradation or hydrolysis, physical dynamic coatings create a continuously changing surface in response to physical stimuli, such as electricity, gas, magnetic fields, or marine flow.^[63–67,68] These coatings not only make it more difficult for fouling organisms to settle but also facilitate the detachment of organisms already adhered to the surface through dynamic deformation.^[37] Therefore, physical dynamic coatings driven by

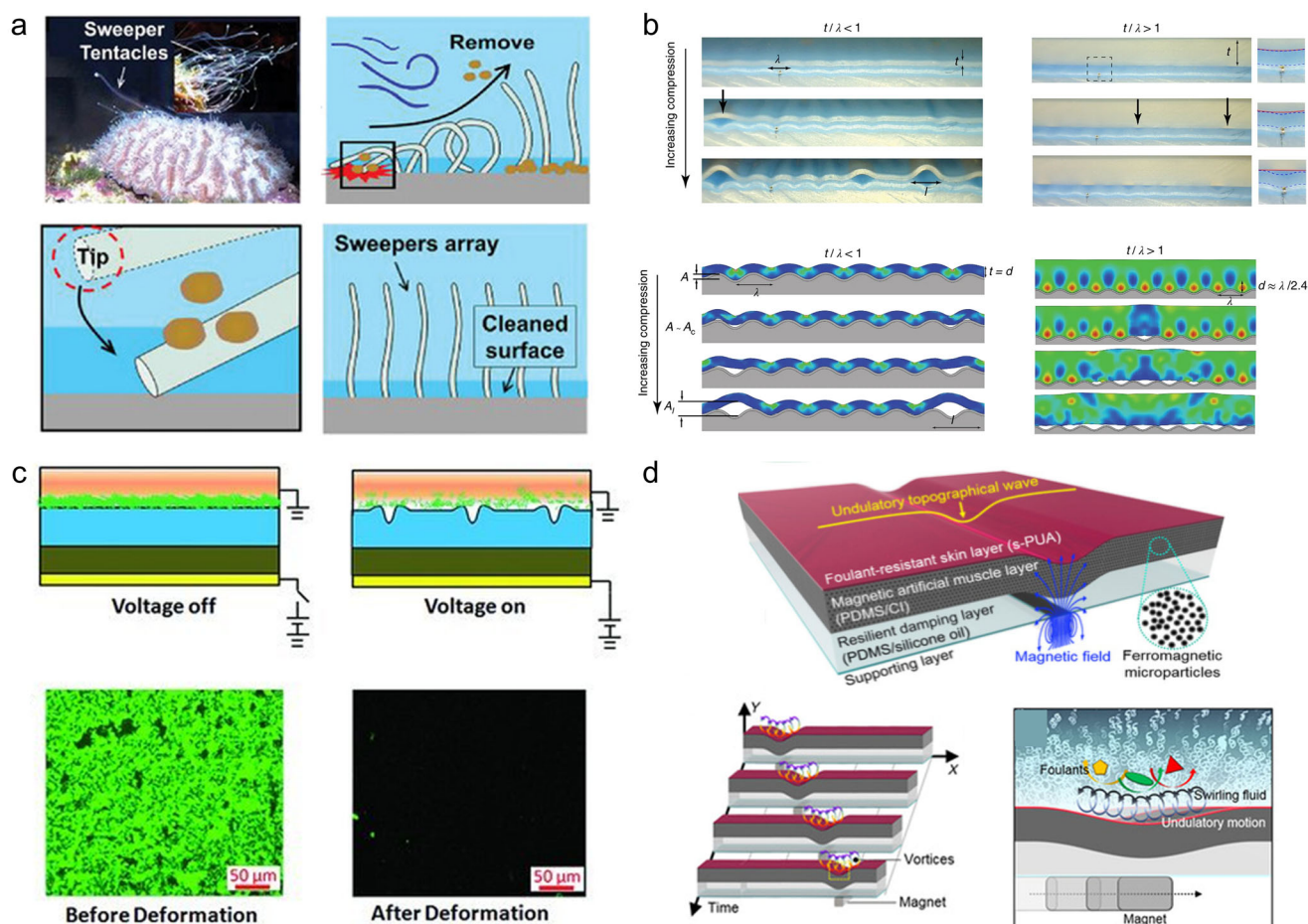


Figure 3. a) Active self-cleaning surfaces inspired by sweeper tentacles of coral. The unit can penetrate the viscous sublayer and remove settled contaminants using energy from the outer turbulent flows. Adapted with permission.^[70] Copyright 2023, Wiley-VCH. b) Representative images of experimental soft silicone patches on wrinkling surfaces and finite-element simulations. Reproduced with permission.^[71] Copyright 2018, Springer Nature. c) Detachment of bacterial biofilms from dielectric elastomers under applied voltages, where over 95% of the biofilm adhered to the elastomer was removed by deformation. Reproduced with permission.^[63] Copyright 2013, Wiley-VCH. d) Schematic of the multilayered dynamic undulatory composite and illustrations showing the propagating undulatory topographical wave along with the translation of the magnet and the resulting topography wave-induced sweeping of foulants. Reproduced with permission.^[64] Copyright 2019, American Association for the Advancement of Science.

external forces hold considerable potential for applications in the marine industry.

Studies have shown that soft surfaces with low elastic modulus can undergo considerable deformation under continuous micro-motions, helping to resist biofouling adhesion.^[37,67,69] Inspired by this concept, Liu et al. fabricated a self-cleaning surface with filament-like sweepers using 3D-printed templates and replica methods (Figure 3a).^[70] Under oscillating flow, the sweepers penetrate the viscous sublayer and remove contaminants by harnessing the energy of turbulent flows. Additionally, the sweeper array can fully clean its coverage area within 10 s through coordinated symplectic wave-like movements. This self-cleaning surface operates via fluid–structure interactions between the sweepers and the flow, introducing a novel approach to antifouling design. Tian et al. developed graphene–silicone elastomer (GSE) composites and incorporated a tentacle structure, creating TS-GSE coatings.^[66] The negatively charged surface of the coating affects bacterial adhesion through electrostatic interactions, while the

tentacle structure undergoes harmonic motion under flow, rendering the elastic surface dynamic and providing fouling-release capabilities. As a result, the TS-GSE coating, with optimized elastic modulus, low surface energy, and strong electronegativity, exhibits effective antifouling performance under both static and dynamic conditions.

Additionally, Pocivavsek et al. introduced a novel physical dynamic antifouling approach based on actuated surface topography.^[71,72] A smooth surface can transform into a wrinkled one owing to elastic instabilities, inducing topography-driven delamination of adhered films. During this process, the surface remains intact, enabling repeated actuation and creating a dynamic surface that prevents foulant accumulation (Figure 3b). This antifouling strategy can be implemented by combining deformable elastomers with external electrical or pneumatic stimuli to produce a physical dynamic surface.^[63,68] For example, Zhao et al. demonstrated that the dynamic surface and topography of elastomers can be actuated by external stimuli, such as electrical or

pneumatic forces, to detach fouling organisms.^[63,68] They fabricated an electro-active antifouling coating using a silicone elastomer, a rigid insulating substrate, and a metal foil. When a direct current voltage was applied to the metal foil, an electric field was generated in the elastomer, causing the surface to deform into a “crater” pattern. Once the electric field was removed, the surface returned to its flat state. This voltage-induced surface deformation effectively detaches adherent biofilms (Figure 3c). Ko et al. developed a magnetically responsive multilayered composite as a physical dynamic antifouling material capable of generating undulatory surface waves with controlled length and time scales.^[64] These undulatory waves produce strong vortices and high shear flows near the surface, effectively sweeping away foulants and preventing their initial attachment (Figure 3d).

Although physical stimuli such as electricity and magnetism have been used to create dynamic coatings, surfaces that become unstable through natural forces without external power offer broader applicability as physical dynamic antifouling coatings. Future development of physical dynamic surfaces may focus on integrating them with other antifouling strategies, such as incorporating fouling-resistant groups, to enhance their antifouling performance, particularly in the absence of external energy to sustain surface instability. Furthermore, materials capable of surface chain reorientation to display disparate surface energies depending on the wetting state have the potential to be used as physical dynamic antifouling coatings. These coatings dynamically alter their surface chemistry at the air-liquid-solid interface and have been shown to significantly reduce biofilm formation under intermittent immersion conditions.^[73,74] For example, Yang et al. prepared amphiphilic copolymer coatings by initiated chemical vapor deposition. The coating surfaces capable of dynamic chain reorientation present zwitterionic groups at the water–solid interface and fluorinated side chains at the air–solid interface, minimizing surface energy under both dry and wet conditions. This surface demonstrated superior performance in preventing biofilm formation at the air–liquid–solid three-phase interface, outperforming its hydrophilic zwitterionic and hydrophobic fluorinated counterparts, emphasizing the advantages of the physical dynamic surface with switchable surface energy.^[75]

3. Fouling-Release Coatings

FRCs provide an effective and economical approach to combating biofouling. They feature low surface energy, low elastic modulus, and smooth surfaces, which weaken the adhesion between marine organisms and the coating, making it easier to remove attached organisms via water shear or mechanical cleaning.^[19] Polymers with low surface energy, such as silicones and fluoropolymers, have been widely investigated for FRC applications.^[76] In particular, silicone elastomers—typically synthesized through hydrosilylation or condensation reactions (Figure 4a)—are the most commonly used FRC materials owing to their eco-friendliness, drag reduction capabilities, and chemical stability. As shown in Figure 4b, fouling organisms settled on the surface can be easily removed by hydrodynamic shear.^[76] Furthermore, the low elastic modulus of silicone elastomers makes the surface deformable, hindering biofouling settlement.^[19] However, traditional silicone-based FRCs often

suffer from limitations such as poor substrate adhesion, low mechanical strength, and limited fouling resistance, which restrict their practical applications (Figure 4c).^[19] Recently, considerable efforts have been made to improve both their mechanical properties and fouling resistance under static conditions. Besides, modeled after the smooth, lubricated surfaces of plants and animals, slippery surfaces that increases the difficulty of fouling adhesion are also the directions of FRCs. This section reviews recent advances in the development of fouling-release coatings.

3.1. Soft Fouling-Release Coatings

High mechanical strength is crucial for soft silicone-based FRCs to prevent damage and extend service life. Physically incorporating nanofillers such as carbon nanotubes (CNTs), graphene oxide sheets, or titanium oxide (TiO₂) can enhance mechanical properties.^[77–79] Chemical modification, for instance by introducing epoxy or urethane/urea groups into silicone, can improve adhesion strength.^[80–82] Combining both approaches, Xie et al. developed an FRC with high tensile strength and enhanced adhesion by incorporating fluorinated nanodiamonds into a poly(dimethylsiloxane) PDMS-based polyurea coating (Figure 5a).^[83] The hydrogen bonds of the urea groups provide strong adhesion strength (Figure 5b). Simultaneously, the fluorinated nanodiamonds enhance mechanical properties and lower surface energy, improving the coating's fouling-release performance. This biocide-free, modified PDMS-based polyurea coating maintained effective fouling-release capability for four months in marine field tests (Figure 5c).

It should be noted that while the approaches described above enhance the mechanical properties of silicone-based FRCs, their fouling-release performance under static conditions remains limited. Therefore, considerable efforts have focused on improving fouling resistance, for example, by incorporating antifoulants or hydrophilic groups that form a hydration layer. Xie et al. developed a self-stratifying silicone coating containing non-leaching antifoulants by grafting telomers of dodecafluoroheptyl methacrylate (DFMA), triclosan acrylate, and 3-mercaptopropyl trimethoxysilane (KH590) onto bis-silanol-terminated PDMS (Figure 5d).^[84] The fluorine drives triclosan to enrich at the surface owing to its low surface energy, inhibiting both marine bacterial biofilm growth and diatom adhesion. Furthermore, the coating possesses low surface energy ($\sim 20 \text{ mJ m}^{-2}$) and a low elastic modulus ($< 2 \text{ MPa}$), resulting in excellent fouling-release performance. In marine field tests, it maintained outstanding antifouling efficacy for over six months without leaching antifoulants (Figure 5e). Other antifoulants, such as Econe or TCPM can also be incorporated to enhance antifouling performance.^[85,86] Additionally, hydrophilic groups—including poly(ethylene glycol) (PEG), zwitterions, and polyvinylpyrrolidone (PVP)—can be introduced into silicone coatings to create amphiphilic surfaces that resist biofouling adhesion.^[87–89] The amphiphilic nanodomains on the surface, exhibiting chemical ambiguity, can reduce the entropic and enthalpic driving forces for marine protein adsorption (Figure 5f).^[24] Krishnan et al. investigated how block copolymer surfaces with amphiphilic side chains influenced marine organism adhesion.^[90] They proposed a mechanism for the reconstruction of ethoxylated fluoroalkyl side chains

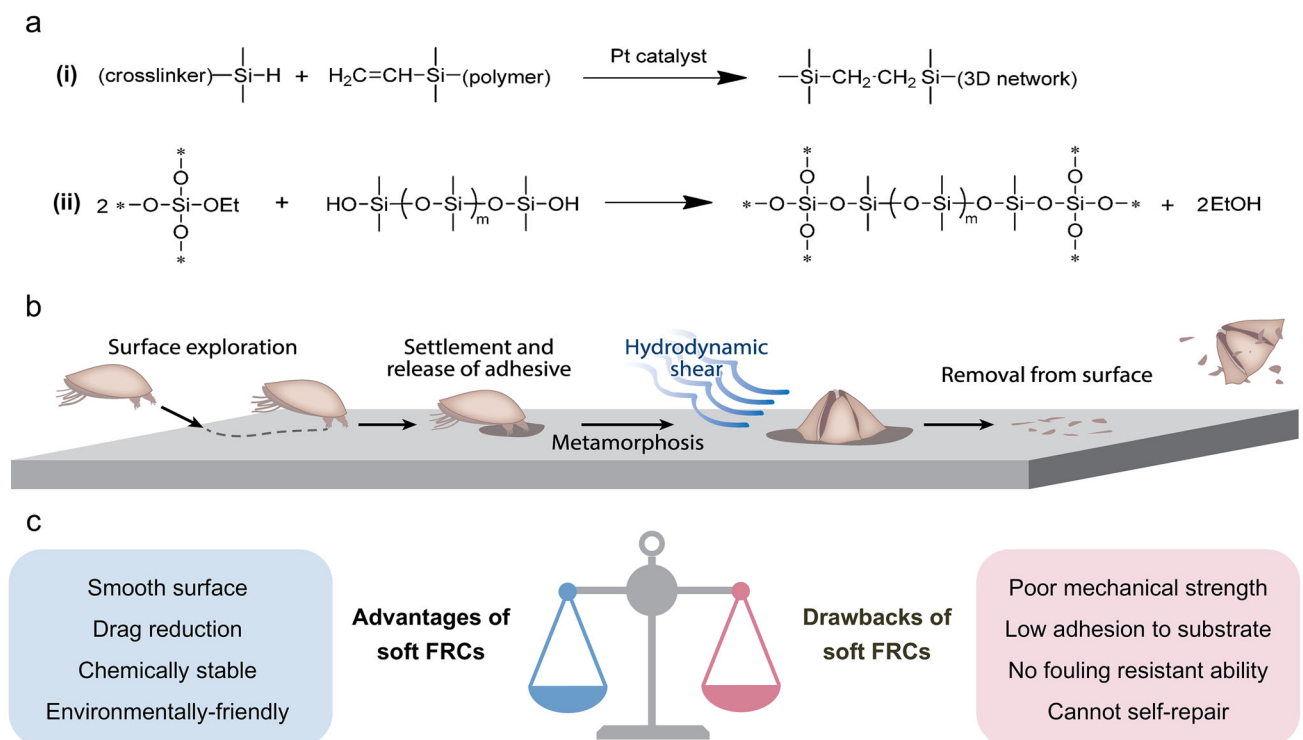


Figure 4. a) Hydrosilylation and condensation reactions for the preparation of silicone elastomers. Reproduced with permission.^[19] Copyright 2020, American Chemical Society. b) Fouling-release coatings present a surface that fouling organisms can settle on but cannot firmly attach to. Here, a barnacle cyprid explores the surface, releases adhesive, and metamorphoses into an adult barnacle. Its attachment is weak; thus, hydrodynamic shear from water jets or a ship's motion easily removes it. Reproduced with permission.^[76] Copyright 2019, Annual Reviews. c) Advantages and drawbacks of soft FRCs.

upon immersion in water. As shown in Figure 5g, the fluoroalkyl side chains orient toward the air, while the PEG chains orient toward the surface upon water immersion. This surface reconstruction renders the coating as hydrophilic as a PEGylated surface, facilitating the removal of *Navicula*, which adheres weakly to hydrophilic surfaces. Zeng et al. grafted telomers of DFMA, poly(ethylene glycol) methyl ether methacrylate (PEGMA), and KH590 onto bis-silanol-terminated silicone to create an FRC with non-leaching amphiphilic side chains, effectively inhibiting marine biofouling under static conditions.^[91] The low surface energy of fluorine drives the amphiphilic telomer to concentrate at the surface, giving the coating low surface energy and low elastic modulus, which provide fouling-release capabilities. Additionally, the PEG segments form a hydration layer on the surface, imparting strong fouling-resistance under static conditions. Guo et al. developed an antifouling coating by incorporating PVP into PDMS to create a triblock amphiphilic copolymer, PVP-PDMS-PVP.^[89] In this system, hydrophobic PDMS segments are anchored in the network, while hydrophilic PVP segments migrate to the surface under water induction. The modified catheters exhibited 96.2% and 95.0% adhesion inhibition against *E. coli* and *S. aureus*, respectively. In addition, zwitterions can be incorporated into silicone to enhance fouling resistance.^[92–94] However, zwitterionic polymers have higher surface energy than PDMS, causing them to remain trapped in the bulk and limiting their antifouling effectiveness. Moreover, highly polar zwitterions are challenging to integrate directly into

coatings owing to incompatibility with nonpolar materials, and their considerable swelling in water can compromise mechanical properties.^[95] To overcome these challenges, Hu et al. synthesized a novel oligomer composed of the zwitterionic monomer tertiary carboxybetaine dodecafluoroheptyl ester ethyl acrylate (TCBF) and 3-mercaptopropyltriethoxysilane, which was then chemically grafted onto bis-silanol PDMS.^[96] During coating formation, the fluorocarbon chains drive the telomer to enrich at the surface, generating zwitterions upon hydrolysis and imparting fouling-resistance. Simultaneously, the hydrophobic TCBF in the bulk reduces PDMS swelling and preserves fouling-release performance. The resulting coating demonstrates strong fouling resistance against both bacteria (*Pseudomonas* sp.) and diatoms (*N. incerta*).

To further extend the service life of FRCs, dynamic bonds can be incorporated to provide self-healing capabilities.^[97,98] Liu et al. developed a self-repairing coating composed of PDMS-based polyurea (PDMS-PUa) and the organic antifoulant DCOIT. Owing to hydrogen bonding from urea groups and the flexibility of PDMS segments, the polymer can self-heal at room temperature. Coupled with the sustained release of DCOIT, the coating demonstrated excellent antifouling performance over six months in marine field tests.^[97] Hu et al. reported a PDMS coating with both self-healing and fouling-resistant properties, achieved via coordination bonds between 2-(2-benzimidazolyl)ethanethiol (BET) and zinc ions.^[99] The cross-linked metal coordination in the coating is reversible, providing effective self-healing at room

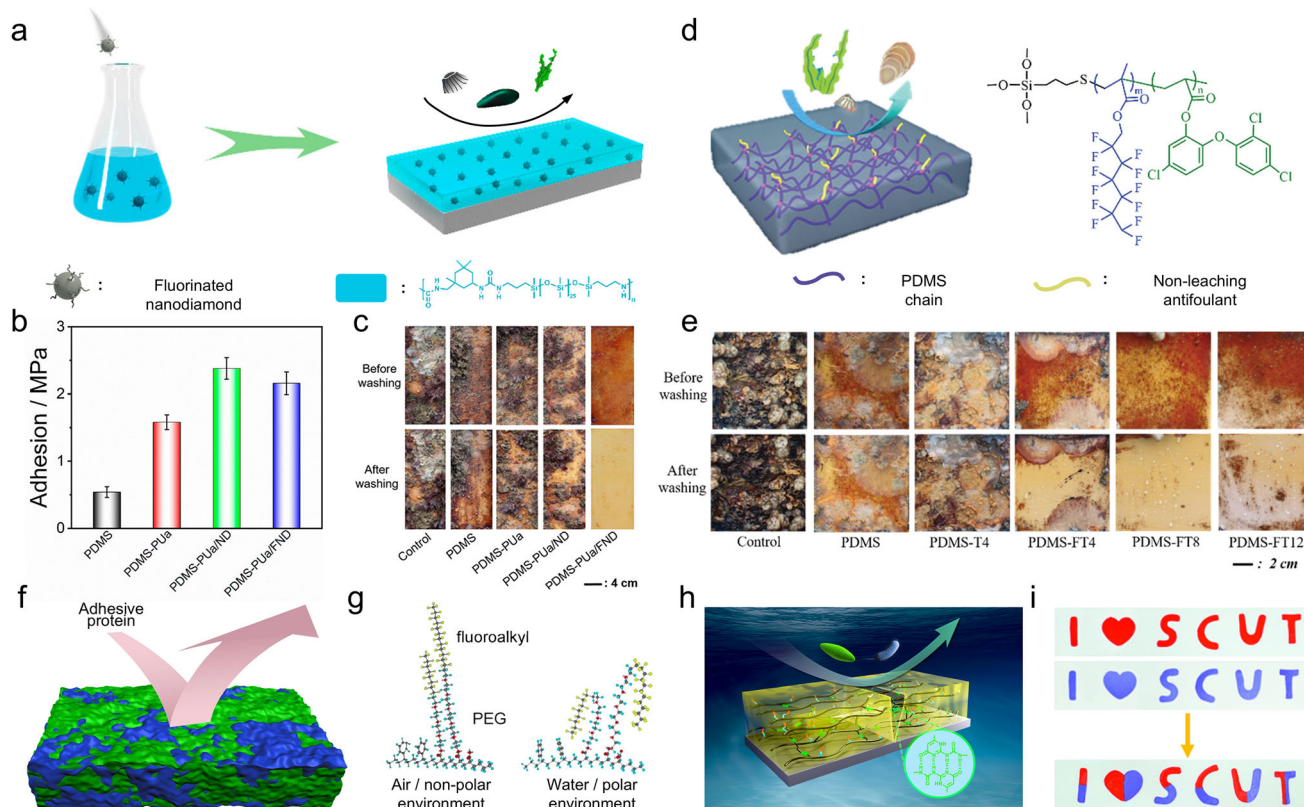


Figure 5. a) Strategy to enhance the adhesion strength of PDMS coatings modified with fluorinated nanodiamonds. Adapted with permission.^[83] Copyright 2020, American Chemical Society. b) Adhesion strength of PDMS-based coatings on an epoxy panel. Reproduced with permission.^[83] Copyright 2020, American Chemical Society. c) Images of PDMS-based coatings after four months of immersion in natural seawater. Adapted with permission.^[83] Copyright 2020, American Chemical Society. d) Strategy to improve the antifouling properties of PDMS coatings modified with nonleaching antifoulants. Adapted with permission.^[84] Copyright 2019, Wiley-VCH. f) Schematic of anti-biofouling via an amphiphilic surface. Reproduced with permission.^[24] Copyright 2011, Springer Nature. g) Proposed mechanism for surface reconstruction of ethoxylated-fluoroalkyl side chains upon immersion in water. The picture on the left indicates the orientation of side chains in air, whereas that on the right shows the effect of water immersion. Reproduced with permission.^[90] Copyright 2006, American Chemical Society. h) Schematic illustration of self-healing PDMS polymer via hydrogen bonding. i) Images of various healed shapes of PDMS-UPy-T15 samples. Healing occurred at room temperature over 48 h. Adapted with permission.^[100] Copyright 2021, The Royal Society of Chemistry.

temperature. Additionally, the zinc–imidazole complexes act as non-leaching antifoulants, enabling strong fouling resistance. Lin et al. incorporated 2-ureido-4[1H]-pyrimidinone (UPy) and amphiphilic pendant chains into a silicone-based polyurethane to enhance both fouling resistance and self-healing capabilities (Figure 5h).^[100] The UPy groups form quadruple hydrogen bonds, imparting excellent self-healing and high elasticity.^[101–103] As shown in Figure 5i, separated polymer segments can recombine and heal into a unified structure. Silicone-based coatings with self-healing capabilities and strong antifouling performance are expected to achieve longer service life in marine environments, while simultaneously reducing chemical usage and waste generation.

3.2. Hard Fouling-Release Coatings

The silicone coatings discussed above are primarily linear chains or cross-linked chains with low densities, resulting in low me-

chanical robustness. Mechanical reinforcement through physical blending of nanofillers can lead to poor interfacial compatibility or filler agglomeration.^[104] To address these issues, stereoscopic polysiloxanes with high crosslink density—referred to as polymer–ceramic hybrid coatings—have been developed (Figure 6a).^[105] These hybrid coatings feature an inner core of inorganic components and an outer layer of reactive organic groups, providing both ceramic-like hardness and polymer-like flexibility (Figure 6b).^[21,105–107,108] Using building blocks of varying sizes, the fabrication strategies and resulting properties of the coatings can be tailored (Figure 6c).^[21,109] Specifically, chemical modification of the organic chains with diverse functional groups can impart additional capabilities—such as anti-liquid adhesion, anti-biofouling, and self-healing—broadening the applications of hybrid coatings in marine industries and beyond (Figure 6d).^[110–116,117]

For instance, Zhang et al. developed a recyclable glassy polymeric material by combining polysilsesquioxane (POSS), isocyanate-terminated polyurethane precursors, and functional

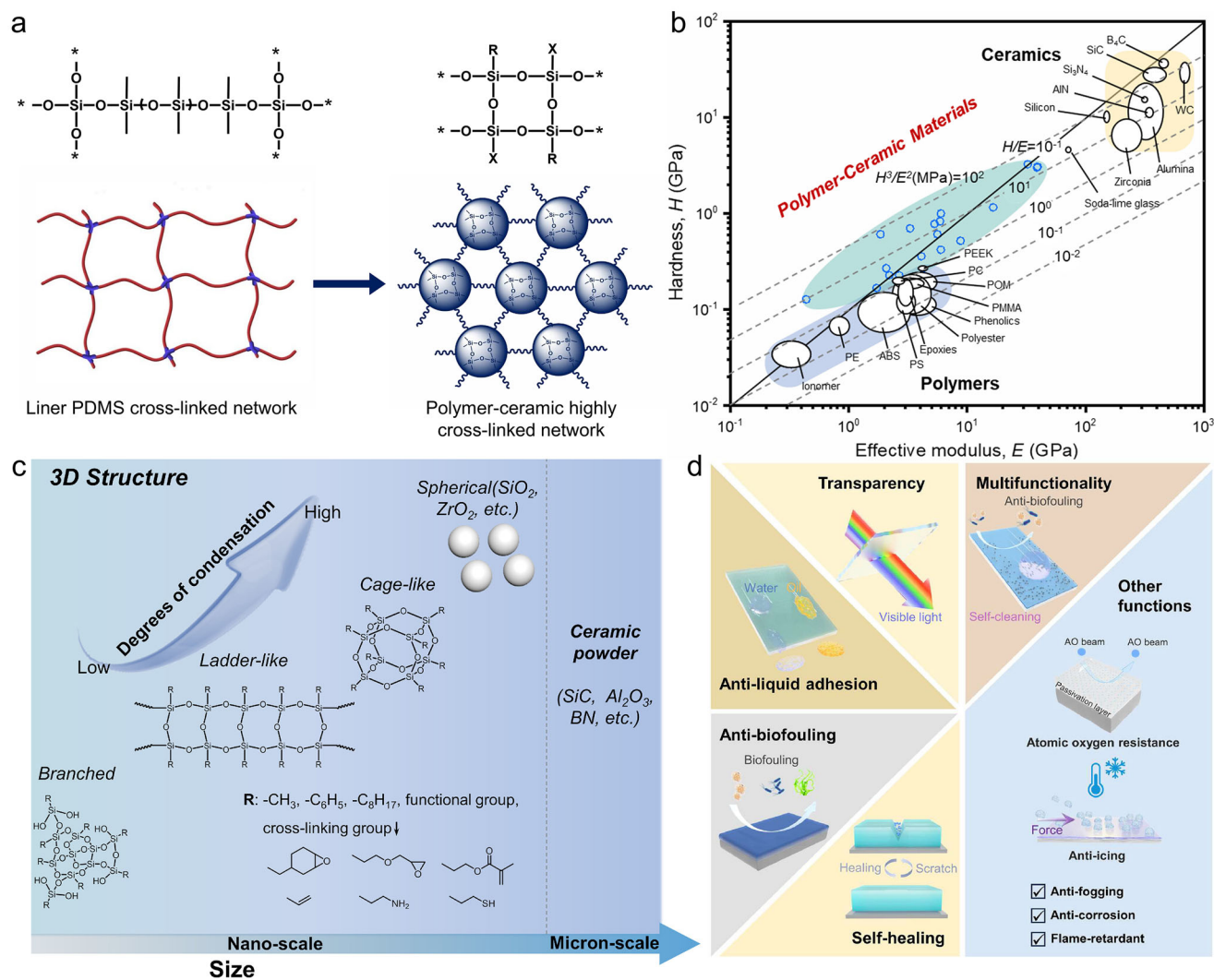


Figure 6. a) Structural schematic of linear PDMS cross-linked network and polymer–ceramic highly cross-linked networks. Adapted with permission.^[110] Copyright 2023, American Chemical Society. b) Classification of polymer–ceramic hybrid coatings. Adapted with permission.^[21] Copyright 2024, Elsevier. c) Summary of the building blocks of polymer–ceramic hybrid coatings. Reproduced with permission.^[21] Copyright 2024, Elsevier. d) Schematic illustration of various functional polymer–ceramic hybrid coatings. Reproduced with permission.^[21] Copyright 2024, Elsevier.

PDMS.^[118] POSS acts as a self-reinforcing agent to provide rigidity, while dynamic thiourethane covalent bonds confer recyclability. The PDMS segments can self-enrich at the surface, imparting anti-liquid adhesion, self-cleaning, and deicing properties. This material, featuring hard reinforcers and soft organic chains, exhibits high hardness (up to 5H), good flexibility, and exceptional wear resistance. Owing to the dynamic thiourethane and hydrogen bonds, it can be recycled through hot-pressing or solvent-based methods. You et al. fabricated an antiadhesion coating with self-healing capabilities using a multi-magnitude dynamic bond synergy strategy.^[119] Amino-rich oligosiloxane with PDMS chains was dynamically linked to an amino-rich oligosiloxane containing UPy moieties via coordination bonds and hierarchical hydrogen bonds. The reversible reactions of the hydrogen bonds and Al–N coordination bonds can be triggered by high temperature and water vapor, allowing the polymer chains to reorganize and imparting self-healing properties. Scratches on the coating

surface were fully repaired after exposure to water vapor, achieving a self-healing efficiency of 99.95%. The coating demonstrates outstanding self-healing efficiency while maintaining high hardness (9H) and flexibility (2 mm bending radius). Additionally, the PDMS segments confer excellent antiadhesive properties.

The low surface energy PDMS chains in the coating provide limited anti-biofouling performance under static conditions, restricting their use in marine applications. To overcome this, PEG or zwitterionic groups are incorporated into polymer–ceramic hybrid coatings, imparting antifouling properties while maintaining mechanical robustness.^[95,112,120,121] Chen et al. developed a waterborne antifouling coating by combining amine-terminated hyperbranched polysiloxane with epoxy–zirconium particles functionalized with zwitterionic chains. The resulting highly cross-linked inorganic–organic network (Zr–O–Si) exhibits both high hardness (7–9 H) and good flexibility (≤ 10 mm bending diameter). The zwitterions in the coating form a

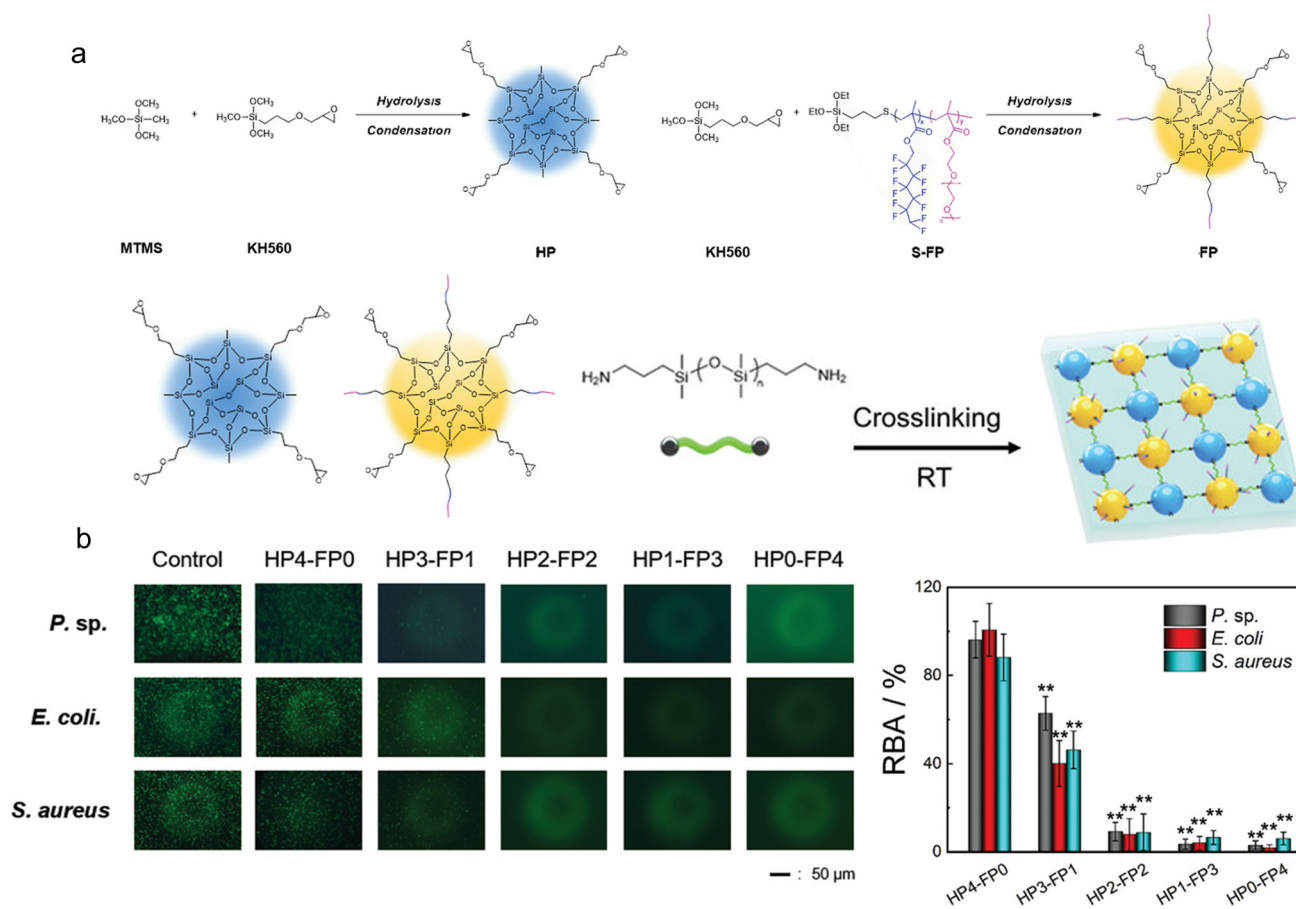


Figure 7. a) Preparation of hyperbranched epoxy-oligosiloxane, fouling-resistant epoxy-oligosiloxane nanoclusters, and their cross-linked network. Adapted with permission.^[121] Copyright 2022, Wiley-VCH. b) Fluorescence microscopy images of *P. sp.*, *E. coli*, and *S. aureus* on the surface of hybrid coatings and the RBA. Reproduced with permission.^[121] Copyright 2022, Wiley-VCH.

hydration layer, providing fouling resistance and excellent underwater oil repellency. However, highly hydrophilic PEG or zwitterionic chains can cause swelling in water, compromising the coating's mechanical properties. To address this, Zhang et al. developed a novel stepwise and self-enrichment strategy to fabricate a highly cross-linked multifunctional hybrid coating with ceramic-like hardness and polymer-like flexibility.^[121] The coating is prepared by combining two types of epoxy-oligosiloxane nanoclusters with amine-terminated curing agents, where one nanocluster imparts hardness and the other provides antifouling properties (Figure 7a). The low surface energy fluorocarbon segments drive the PEG chains to enrich at the coating surface, preventing swelling in water, while the PEG forms a hydration layer that imparts fouling resistance (Figure 7b). Simultaneously, the low-surface-tension PDMS acts as a liquid lubricant, providing self-cleaning and anti-liquid properties.

Polymer-ceramic hybrid antifouling coatings, combining organic functional chains such as PDMS, PEG, or zwitterions with an inorganic network, simultaneously achieve enhanced mechanical strength and excellent antifouling performance. Future research should focus on environmentally friendly cross-linked networks that are self-healing or recyclable, addressing societal

concerns over the environmental and economic impacts of fossil fuel-based polymers.^[122] However, the high crosslink density limits the reversibility of these coatings, making it challenging to develop sustainable polymer-ceramic hybrid marine antifouling coatings for broader applications in marine industries and beyond.

3.3. Slippery Fouling-Release Coatings

In addition to the fouling-release coatings mentioned above, which resist fouling primarily through low surface energy, there is another emerging type of fouling-release coating with a slippery surface that increases the difficulty of fouling adhesion. This type can also be categorized into biomimetic coatings. For instance, inspired by the *Nepenthes* pitcher plant, slippery liquid-infused porous surfaces (SLIPS), which feature an oil layer serving as a physical barrier, have demonstrated excellent fouling-release performance owing to their poor adhesion to biofilms (Figure 8a,b).^[123–125] Moreover, SLIPS with lubricating liquids locked in micro- or nanoporous substrates can repel various liquids, rapidly restore liquid repellency after physical damage, resist ice and pollutant adhesion, and maintain pressure

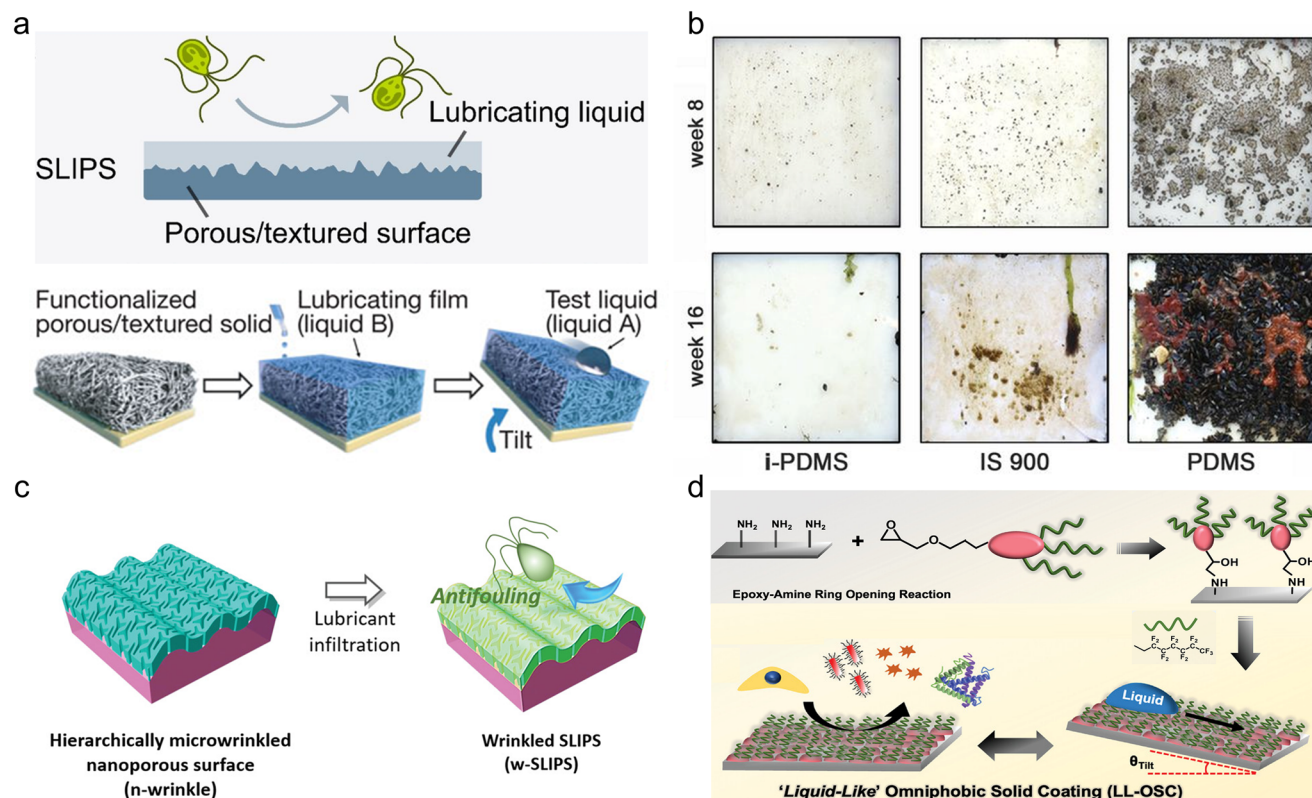


Figure 8. a) Structures of SLIPS and the preparation of SLIPS. Reproduced with permission.^[124] Copyright 2011, Springer Nature. b) Antifouling performance of i-PDMS, PDMS, and IS900 panels in the field. Reproduced with permission.^[125] Copyright 2017, American Association for the Advancement of Science. c) Illustration of the fabrication of a SLIPS with anti-biofouling property. Reproduced with permission.^[128] Copyright 2021, Wiley-VCH. d) Schematic of "liquid-like" omniphobic solid coating (LL-OSC) that exhibited liquid repellence characteristics along with anti-biofouling behavior. Reproduced with permission.^[137] Copyright 2024, Wiley-VCH.

stability.^[124,126] Owing to these advantages, SLIPS hold great potential for marine antifouling applications.^[125]

To maintain the lubricating liquid stably in the substrate, micro/nanotextured and rough substrates with high surface area are required.^[127] Inspired by frog skin, Hsueh et al. fabricated a wrinkled slippery coating by combining surface wrinkling with diblock copolymer (di-BCP) hydrolytic degradation.^[128] They used polystyrene-*b*-polylactide (PS-*b*-PLA) as the core material to form co-continuous PLA networks in a PS matrix after thermal treatment or solvent annealing. Subsequent hydrolysis of the PLA block generated a micro-wrinkled PS matrix containing networked nanochannels. These nanochannels act as lubricant reservoirs, while the micro-wrinkled topography creates curved surfaces that resist biofouling through surface geometry (Figure 8c). This wrinkled slippery coating demonstrates excellent antifouling, anti-icing, and self-cleaning performance. Li et al. developed a SLIPS coating by combining corrosion-protective biofilm elements with pitcher plant-inspired structures. They arranged microbial cells into highly patterned surfaces and then deposited hydrophobic lubricant onto the engineered biofilms. The resulting coating, featuring both a slippery surface and extracellular polymeric substance biopolymers, exhibits broad-spectrum resistance against corrosive media, proteins, bacteria, and algae under complex biological conditions.^[129] However, the lubrication layer can be lost un-

der applied shear stress, which limits the long-term durability of SLIPS coatings.^[130]

To address these limitations, one strategy is to dynamically regulate the surface lubricity to minimize lubricant loss. For instance, Tong et al. incorporated a light- and heat-responsive supramolecular interaction between azobenzene (Azo) and α -cyclodextrin (α -CD) into SLIPS.^[131] Under visible light or heating, *cis*-Azo converts to *trans*-Azo and forms a complex with α -CD, inducing polymer chain shrinkage that squeezes stored lubricant to the surface, thereby modulating surface lubricity in response to environmental changes. Additionally, to enhance antifouling performance under static conditions and improve long-term durability, SLIPS with responsive antifoulant release capabilities have been developed.^[132,133] Tong et al. developed a SLIPS with responsive antifouling properties by integrating ZIF-8 loaded with the eco-friendly antifoulant tralopyril (4-mo-2-(4-chlorophenyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile, Tra), a UV-responsive agent (2-nitrobenzaldehyde, NBA), and a reactive polysiloxane. Under UV irradiation, ZIF-8 (ZIF@TraNBA) degrades, releasing Tra in a controlled manner to enhance the antifouling effect. This smart coating, with adaptive antifoulant release capability, can adjust to varying antifouling demands, thereby improving durability. It should be noted that marine organisms often use multiple antifouling strategies simultaneously. A representative example is soft corals, which

use soft skin for fouling release, secrete natural antifoulants for fouling inhibition, possess soft tentacles for physical dynamic antifouling, and shed slime for chemical dynamic antifouling.^[134] Inspired by such multifunctionality, bioinspired coatings that integrate several strategies can achieve enhanced antifouling performance and durability. For example, inspired by the venomous, sweeping tentacles of sea anemones, biomimetic cilia actuated by magnetic fields have also been incorporated into SLIPS coatings, conferring additional physical antiadhesive properties.^[135] The magnetically actuated rotation of the cilia, combined with a lubricating layer, physically prevents biofouling from adhering to the surface, while the photo-stimulated release of antifoulants enhances antifouling performance under static conditions. This hybrid system exhibits highly responsive anti-biofouling properties, demonstrating considerable potential for marine industry applications.

Another approach to improving the durability of slippery surfaces is the development of liquid-like slippery surfaces that do not rely on infused lubricants.^[136] For instance, Shome et al. prepared a solid slippery coating with high mechanical robustness and long-term stability by covalently grafting fluorinated silanes onto the substrate (Figure 8d).^[137] A mixture of perfluorooctyl triethoxysilane (PFOTS) and (3-glycidyloxypropyl)trimethoxysilane (GPTMS) was hydrolyzed and condensed to form an interconnected silane network. The epoxy-functionalized silane provides optimal spacing for the mobility of the fluorinated chains while ensuring strong anchorage to an amine-prefunctionalized surface, yielding a durable, lubricant-like slippery coating. The liquid-like omniphobic solid coating, capable of repelling various liquids as well as resisting biofilm, protein, and cell adhesion, shows promise as a marine antifouling coating. To further enhance its antifouling performance under static conditions, eugenol encapsulated in cyclodextrins has been incorporated into the polymer chains, enabling the controlled, slow release of natural antifouling agents.^[138]

4. Fouling-Resistant Coatings

Marine biofouling typically initiates with protein adsorption, followed by fouling organisms secreting protein-based adhesives to attach to surfaces.^[139,140] Fouling-resistant coatings can mitigate biofouling by preventing protein adsorption.^[18] Common fouling-resistant materials, including PEG, zwitterionic polymers, and hydrogels, form a hydration layer that acts as a physical barrier to proteins via hydrogen bonding or electrostatic interactions.^[58,141,142] Other types of fouling-resistant strategies—including electrostatic-barrier-based resistance (protein-based coatings), charge-barrier-based resistance (electrical methods), and topographical resistance (biomimetic microstructures)—are being actively explored. Considerable efforts have been made to explore the potential of fouling-resistant coatings for marine antifouling applications.

4.1. Hydration-Based Fouling-Resistant Coatings

Grafting linear PEG brushes onto surfaces is a common strategy to resist protein adsorption owing to their water solubility, biocompatibility, and non-toxicity.^[143,144] Vahabi et al. developed slippery hydrophilic surfaces by covalently attaching PEG

brushes to smooth substrates (Figure 9a).^[145] Owing to the hydration layer and steric hindrance provided by PEG brushes, the surfaces demonstrated excellent fouling resistance by inhibiting fibrinogen adsorption. However, grafting PEG onto various substrates involves complex surface chemistry, making it costly and difficult to scale up for industrial applications, which poses a considerable barrier to its use in marine coatings.^[146,147] To address this, Ma et al. incorporated PEG into polyurethane as a soft segment and studied the resulting protein adsorption behavior on the surfaces. The coating demonstrated excellent protein resistance owing to the hydration layer of PEG segments.^[148] To evaluate antifouling performance under real marine conditions, they performed the first field experiments on anti-protein adsorption materials. The tests showed effective antifouling for four weeks; however, prolonged immersion led to loss of antifouling ability, likely owing to coverage by dead marine organisms and marine debris in the complex marine environment.^[149] To further enhance antifouling performance, degradable polymers with fouling-resistant properties have been developed.^[59] For example, a copolymer with poly(ethylene oxide-co-ethylene carbonate) side chains was synthesized and tested in marine environments, demonstrating improved antifouling on the coating surface.^[149,150] Additionally, Ai et al. developed a hyperbranched polymer containing diethylene glycol units by copolymerizing MDO, vinyl acetate (VAc), and diethylene glycol divinyl ether. The polymer degrades into small fragments via cleavable ester bonds in the main chain, while the hydrophilic diethylene glycol units inhibit marine biofouling adhesion on the surface. The hyperbranched polymer coating, featuring a chemically dynamic surface and fouling-resistant properties, shows promising potential for marine antifouling applications.^[151]

Zwitterionic polymers, which bind water through ionic solvation, represent a new generation of fouling-resistant materials with potential applications in marine antifouling.^[152,153] However, their high hydrophilicity often results in insufficient mechanical strength, limiting practical use.^[154] To address this, HIZs have been developed.^[50,55,155] For instance, Hung et al. synthesized a zwitterionic precursor that generates zwitterions via simple chemical triggers, producing a polymeric elastomer with high bulk mechanical strength and strong surface fouling resistance (Figure 9b).^[155] Xie et al. prepared a degradable polymer with HIZs through the copolymerization of MDO, tertiary carboxybetaine ester (TCB), and 7-methacryloyloxy-4-methylcoumarin (MAMC). With the degradation of the polymer main chains, the zwitterions are continuously regenerated, providing the surface with persistent protein resistance. Because hydrolysis occurs primarily at the surface, the resulting hydrophilic layer is continuously renewed while preventing water from penetrating into the bulk, allowing the coating to remain hydrophobic and mechanically durable.^[155] Additionally, HIZs can be incorporated into FRCs. For example, Huang et al. synthesized a telomer (FT) containing a low-surface-energy fluorine motif and HIZs.^[156] Incorporating FT into polysilazane yielded a hard yet flexible, self-adaptive zwitterionic polysilazane coating with broad-spectrum antiadhesion properties, including resistance to biofouling, liquid adhesion, and scaling. During coating formation, the fluorine motif drives the telomer to migrate to the surface, forming a zwitterionic layer upon pre-hydrolysis. In air, the fluorine motif orients toward the surface, providing

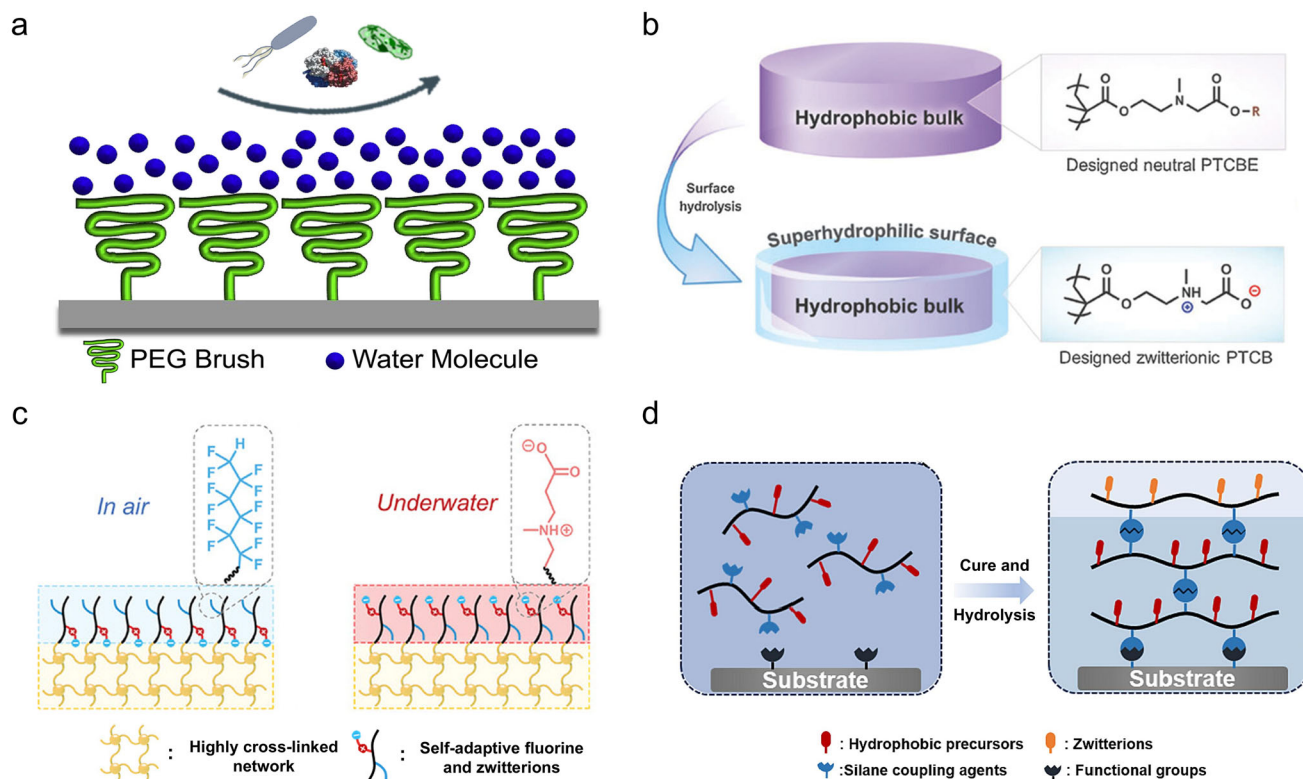


Figure 9. a) Schematic depicting negligible penetration of water molecules between PEG brushes at the critical grafting density. Adapted with permission.^[145] Copyright 2022, Elsevier. b) Hydrolysis of tertiary carboxybetaine ester polymer (PTCBE). Reproduced with permission.^[155] Copyright 2017, Wiley-VCH. c) Schematic of the preparation of self-adaptive zwitterionic polysilazane coatings. Reproduced with permission.^[95] Copyright 2024, Wiley-VCH. d) Schematic of self-generation of the hydrogel layer: the top layer hydrolyzes and the bottom layer bonds to the substrate surface for improved adhesion. Reproduced with permission.^[163] Copyright 2024, Elsevier.

anti-liquid adhesion, whereas underwater, the zwitterions orient outward, imparting anti-biofouling, anti-liquid adhesion, and anti-scaling capabilities (Figure 9c). This mechanically robust zwitterionic polysilazane coating with broad-spectrum antiadhesive properties has potential applications in marine industries, optical devices, pipeline transportation, and other fields.

Hydrogels are 3D cross-linked networks of hydrophilic polymer chains that contain abundant water and exhibit low elasticity, providing excellent fouling-resistant capabilities in aquatic environments.^[141,157,158] Similar defense mechanism can be found in nature. For example, the skin mucus of fish—a natural hydrogel—protects them from invading pathogens owing to its soft, hydrophilic, and gel-like properties in water.^[159] Inspired by the concept, Xue et al. developed a hydrogel-based antifouling coating by incorporating sodium polyacrylate (PAAS) powders into an organic silicon resin.^[160] The PAAS absorbs water and swells to form a hydrogel layer on the surface, preventing marine biofouling. Similarly, Yu et al. reported a hydrogel-anchored metal-organic framework (MOF) coating composed of a biopolymer-based hydrogel and densely packed MOFs, which demonstrates broad-spectrum antifouling performance and long-term protection against microbiologically influenced corrosion.^[161] To be effective in marine antifouling applications, hydrogel coatings require strong adhesion to substrates. Xie et al. developed a self-generating hydrogel coating by copoly-

merizing MMA, acrylic acid, TBSM, and a polyfunctional axiridine cross-linker. The TBSM undergoes hydrolysis in seawater, forming a thin, soft, and dynamic hydrogel layer at the water-contacting surface.^[162] Building on this approach, Zhang et al. designed a self-generating zwitterionic Janus hydrogel paint with strong adhesion, composed of MMA, tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSi), and 3-(methacryloxypropyl) trimethoxysilane (KH570), which further enhances fouling resistance.^[163] The hydrophobic MMA in the paint enhances mechanical performance, while the zwitterionic precursor TCBSi generates a surface hydration layer after hydrolysis, providing antifouling properties (Figure 9d). Notably, the unhydrolyzed hydrophobic bottom layer anchors the coating to the substrate through hydrophobic interactions and chemical bonds from the silane coupling agents.^[163] This design yields higher adhesion strength than conventional zwitterionic hydrogel coatings and delivers excellent resistance to proteins, marine biofouling, and oil contamination owing to the robust surface hydration layer.

Owing to the complex marine environment, antifouling coatings without self-regeneration gradually lose their fouling-resistant ability as inorganic deposits or dead organisms accumulate.^[50] Additionally, they do not provide an extra physical barrier to inhibit corrosion.^[164] To address this, Zhang et al. developed a self-regenerating zwitterionic hydrogel coating

composed of a self-crosslinking degradable polymer with HIZs and a corrosion inhibitor (benzotriazole, BTA) loaded into a MOF.^[156] The degradable polymer can self-crosslink and simultaneously crosslink with the MOF, forming a network that chemically bonds to the substrate, yielding strong adhesion. Through the degradation and hydrolysis of the coating, the zwitterionic surface is continuously renewed, providing sustained fouling resistance against marine biofouling. Concurrently, BTA is released during this process, effectively inhibiting surface corrosion. This approach integrates self-regeneration, antifouling, anticorrosion, and strong adhesion into a single coating, expanding the potential applications of fouling-resistant coatings in marine industries.

For hydration-based fouling-resistant coatings, antifouling efficacy is mainly derived from their hydrophilic nature, which can compromise mechanical strength. Future developments will focus on balancing hydrophilicity with durability. One promising strategy is to combine hydrophobic cross-linked networks with self-enrichment mechanisms, achieving an optimal compromise between antifouling performance and mechanical robustness.

4.2. Other Fouling-Resistant Coatings

Besides common hydration-based fouling-resistant coatings, there are many other types of fouling-resistant strategies, such as electrostatic-barrier-based resistance (protein-based coatings), charge-barrier-based resistance (electrical methods), and topographical resistance (biomimetic microstructures).

Protein-based coatings have been reported to resist fouling adhesion through mechanisms such as electrostatic repulsion, steric stabilization, hydration layer, and the creation of unfavorable conditions for aerobic macrofoulers.^[165–167,168] These properties make them strong candidates for antifouling coatings. For example, Ma et al. investigated the adsorption of proteins such as fibrinogen, bovine serum albumin (BSA), and lysozyme on the surface of soy protein film. When the pH is outside the range between the isoelectric point of soy protein and that of the foreign protein, soy protein generally resists the adsorption of the foreign protein due to electrostatic repulsion.^[165] Furthermore, Pillai et al. demonstrated that protein extracted from fish can significantly reduce the nonspecific adsorption of fibrinogen and albumin on various surfaces due to steric repulsion.^[165] Being the most abundant protein in plasma, bovine serum albumin (BSA) has demonstrated both surface-independent anchoring capability and antifouling properties owing to its amphiphilic nature.^[169–171] The hydrophilic regions of BSA can bind surrounding water molecules via electrostatic interactions, forming a hydration layer that prevents biofouling adhesion, while the other regions facilitate substrate-independent anchoring through multiple molecular interactions.^[172–174,175] For instance, Hu et al. developed an antifouling coating via a one-step aqueous supramolecular assembly of BSA.^[167] Using tris(2-carboxyethyl)phosphine (TCEP) to reduce intramolecular disulfide bonds, BSA aggregated into amyloid-like structures, forming a dense proteinaceous nanofilm. The proteinaceous nanofilm exhibits durable antifouling performance, effectively preventing biofouling adhesion even after ultrasonic treatment. To overcome the inherent limitations of native protein coatings—where the

charge properties of BSA are influenced by complex fluid conditions, reducing interfacial hydration and diminishing antifouling performance in practical applications—zwitterionic sulfobetaine methacrylate (SBMA) segments, which are more stable to pH and salinity changes in aqueous solutions than proteins, have been grafted onto the native BSA molecules.^[176–178] The resulting bionically engineered protein coating demonstrated excellent antifouling performance, as evidenced by reversible adsorption–desorption tests with milk. Xiang et al. developed a protein-derived coating with high mechanical strength and strong hydration capacity using a dual-functional metal coordination strategy.^[179] The coating exhibits excellent lubricity, superior wear resistance under high contact pressure, and robust antifouling capability against various biofouling species. Besides, Wang et al. developed a tenacious and stable coating composed of ovalbumin and tannic acid, which adheres strongly to hydrophobic polymer materials and effectively repels bacterial adhesion. By incorporating silver ions that are reduced in situ, the coating gains broad-spectrum antibacterial properties. Such coating shows great potential for application in marine antibiofouling.^[180] Furthermore, Lindgren et al. proposed a novel, non-toxic approach to combat biofouling on submerged surfaces by depleting oxygen through microbial degradation of the protein component, creating unfavorable conditions for aerobic macrofoulers. Field exposure of panels over sixteen weeks demonstrated that protein-rich coating surfaces reduced fouling by barnacles and bryozoans by 80% and nearly 100%, respectively. These results suggest that this innovative technology could serve as a non-toxic alternative to copper-based antifouling paints, particularly for pleasure boats in sensitive environments.^[168] However, despite the promising performance of protein-derived coatings, their long-term effectiveness in marine environments requires further investigation.

The electrical method is an alternative approach for designing fouling-resistant coatings. It has been reported that applying a surface potential to an electrically conducting surface can prevent biofouling adhesion.^[181,182] However, this charge-barrier-based resistance approach requires an external power supply and poses a risk of electrochemical corrosion owing to water exposure of the conductive surfaces. To overcome these limitations, Zhao et al. developed an integrated triboelectric wave harvester (I-TEWH), which generates electrical oscillations to prevent biofouling adhesion without the need for an external power source.^[183] This strategy effectively combats a variety of biofouling organisms and provides continuous protection by harnessing ambient wave energy. Similarly, Sun et al. developed wind-driven triboelectric nanogenerators (TEGs) that generate high-pulse electrical fields capable of inducing bacterial cell membrane damage.^[184] Furthermore, the generation of reactive oxygen species enables a high water sterilization rate (99.99%) and efficient degradation of organic pollutants (>95%) within 20 minutes. Gaw et al. applied low-voltage square-waveform pulses to a surface to generate a continuous gas bubble layer via electrochemical methods, effectively preventing bacterial adhesion and growth.^[185]

Microstructured surfaces developed by animals and plants play vital roles in their survival. For example, the microstructures of lotus leaves impart excellent self-cleaning properties.^[186] The microscale structures on shark skin help reduce drag during swimming.^[187] The hierarchical structures of gecko feet

enable easy attachment to and detachment from surfaces.^[188] Inspired by these natural strategies, extensive research has been performed on biomimetic microstructured surfaces with fouling-resistant properties.^[1,189] The underlying mechanism of topographical resistance still remains unclear. Changes in surface wettability resulting from topography may contribute to this resistance.^[190] Additionally, microtopographies could influence near-surface fluid dynamics, thereby microhydrodynamically preventing settlement.^[191] Nevertheless, the most developed antifouling mechanism of microstructured surfaces may involve providing fewer attachment points than are required by fouling organisms, thereby resisting fouling adhesion.^[192] For example, Petronis et al. found that microtextured PDMS surfaces can prevent barnacle settlement and concluded that the inclination of the topography elements are important features for macrofouling resistance.^[193] In addition, some microstructured surfaces can exhibit mechano-bactericidal activity, causing morphological damage to bacterial cells and demonstrating antifouling performance. Ivanova et al. investigated the influence of cicada wing microstructure on *Pseudomonas aeruginosa*.^[194] According to scanning electron micrographs of the cicada wing surface, the wings are covered with a periodic topography consisting of hexagonal arrays of nanoscale pillars. When bacteria adhere to the wing surface, the cells appear to be penetrated by the nanopillars, and subsequent viability analysis revealed that these penetrated cells were, in fact, dead. Combining the above two features, Wang et al. also designed ZnO/Al₂O₃ core-shell nanowire architectures with tunable geometries to study the influence of microstructure on marine biofouling.^[195] They found that the hydrophilic nanowires reduced fouling coverage by decreasing the effective settlement area and mechanically penetrating the attached cells. It should be noted that biomimetic coatings with microstructures (e.g., shark-skin- or lotus-inspired structures) demonstrate excellent topographical resistance properties, and hold considerable potential for marine applications. However, the high costs and limited scalability of fabricating such microstructures hinder widespread industrial use.^[189] Future research should focus on developing simple, cost-effective methods for generating microstructured coatings.

5. Emerging Antifouling Strategies

In addition to conventional antifouling coatings, emerging strategies—including fouling-defense strategies (natural antifoulants, enzymatic antifouling, UV radiation), passive and active cleaning technology (polymer cleaning technology and underwater robotics)—are being actively explored. This section summarizes the latest advances in these approaches and discusses potential directions for their future development.

In nature, many organisms have evolved ingenious fouling-defense strategies and developing antifouling coatings by mimicking these chemical defenses has attracted considerable interest.^[37,38] It has been reported that many natural species secrete antifouling substances to defend against other organisms (Figure 10a).^[37] For instance, marine invertebrates such as corals can produce mucus to protect themselves from fouling microorganisms.^[196] For marine plants, Nys et al. reported that secondary metabolites from red algae exhibited antifouling activity against barnacles, macroalgae, and marine bacteria.^[197]

Natural products from marine microorganisms also exhibit biofilm-inhibiting properties and hold great potential as marine antifoulants.^[198] For example, Qian et al. identified 5-octylfuran-2(5H)-one from *Streptomyces* species, a marine bacterium, as an ecologically benign antifoulant.^[199] Qi et al. developed novel antifouling macrodiolides from the deep-sea-derived fungus *Trichobotrys effuse* DFFSCS021, which exhibited strong activity against the settlement of *Bugula neritina* and *Balanus amphitrite* larvae.^[200] Several chemical compounds from terrestrial plants also demonstrate antifouling properties. For instance, capsaicin, derived from chili peppers, shows excellent antifouling activity while maintaining low environmental toxicity.^[201] Feng et al. reported that the plant alkaloid camptothecin had potential as an antifoulant.^[202] Additionally, paeonol, another plant-derived compound, shows promise as a natural antimicrobial and anti-biofilm agent.^[203] Based on these findings, developing natural antifoulants derived from diverse organisms represents a promising strategy to combat biofouling.^[37,38]

However, the extraction of natural antifoulants is often complex and costly, limiting their commercial viability. Synthetic analogs offer a promising alternative to overcome these challenges. Combining molecular and biochemical methods to study the biosynthesis, production, and dynamics of signals in marine biofilms can drive effective development of marine antifouling strategies.^[204] For instance, Qian et al. analyzed the structure–activity relationships of antifouling compounds isolated from marine bacteria and designed a simplified butenolide derivative that retains excellent antifouling activity, providing an effective and non-toxic solution.^[199,205] Moreover, the release rate of antifoulants is a critical factor in determining the long-term stability and effectiveness of antifouling coatings. By incorporating antifoulants into suitable polymer matrices, their release rate can be accurately regulated, improving the long-term efficacy of marine antifouling coatings. For example, Ma et al. combined chemical dynamic surfaces with natural antifoulants by developing biodegradable polymer coatings loaded with butenolide. The butenolide is gradually released from the polymer, providing effective antifouling activity for at least three months.^[199,206] Following the “From the nature, for the nature” concept, Pan et al. developed an eco-friendly coating composed of bio-derived poly(lactic acid) (PLA)-based polyurethane and the natural antifoulant butanolide (Figure 10b).^[207] Because PLA is typically sourced from crops, both the polymer binder and the antifoulant are renewable and environmentally friendly. This coating exhibits strong antifouling performance and holds promise for high-performance, copper-free marine coatings.

In addition to deriving inspiration from natural bioactive molecules, the antifouling mechanisms of marine species can also guide the design of novel antifoulants. For instance, inspired by the haloperoxidase (HPO) enzymes secreted by marine algae, Tremel et al. demonstrated that vanadium pentoxide (V₂O₅) nanowires could inhibit marine biofouling without harming marine organisms.^[208] These V₂O₅ nanowires, which exhibit peroxidase-like activity, are stable, inexpensive, and readily available as catalysts. They catalyze the reaction of hydrogen peroxide with bromide to generate small amounts of hypobromous acid, a compound highly toxic to microorganisms, thereby providing an effective antibacterial effect.^[209,210] As shown in Figure 10c, the V₂O₅ nanowires could inhibit biofouling on surfaces. However,

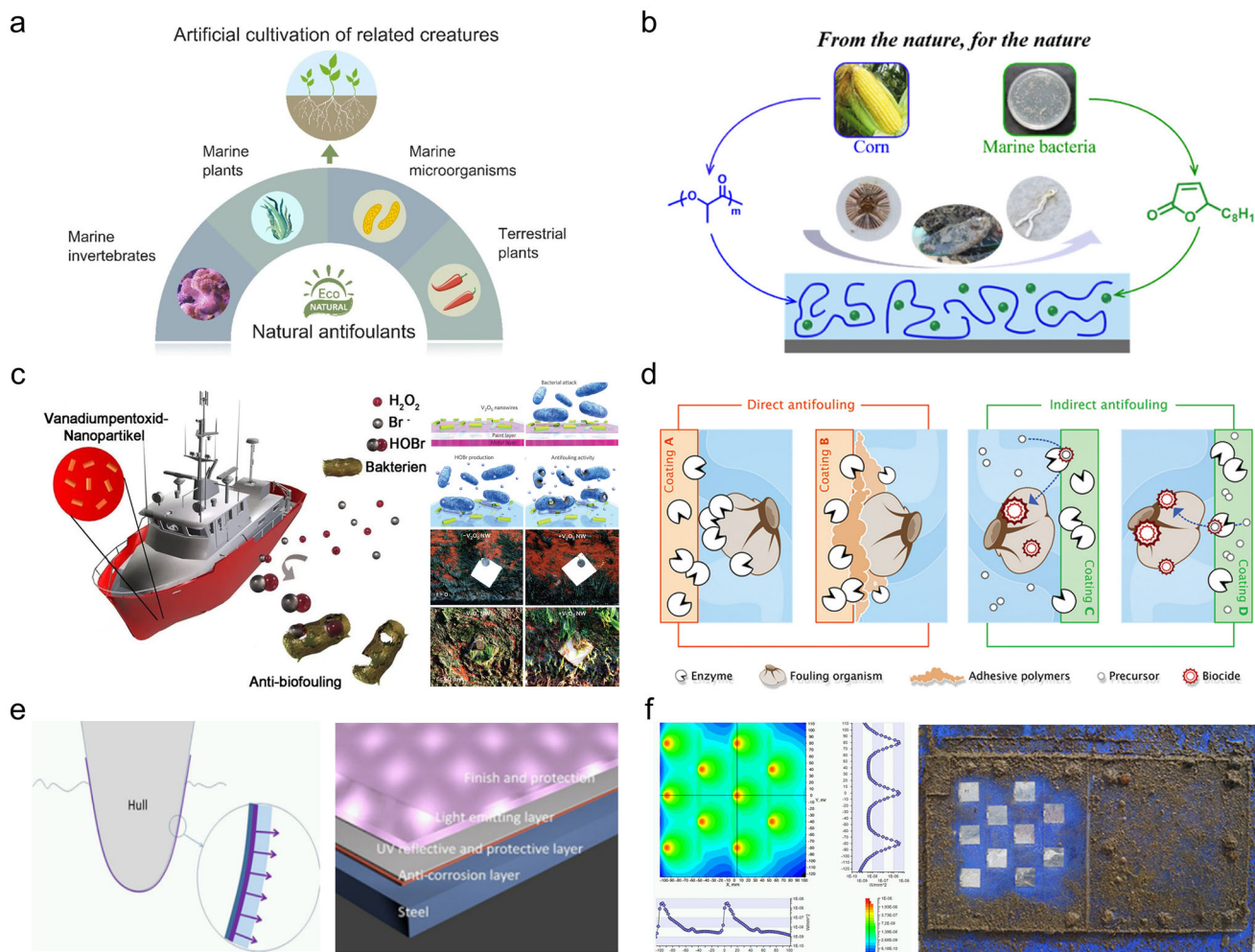


Figure 10. a) Schematic illustration of the development of natural antifoulants. Reproduced with permission.^[37] Copyright 2022, Elsevier. b) Anti-biofouling mechanism of ecofriendly coating with PLA-based polyurethane and natural antifoulant. Adapted with permission.^[207] Copyright 2020, American Chemical Society. c) Bactericidal properties of V_2O_5 nanowires. Reproduced with permission.^[208] Copyright 2012, Springer Nature. d) Classification and proposed mechanism of enzymatic antifouling. Coating A is based on biocidal direct antifouling. Coating B is based on adhesive degrading direct antifouling. Coating C is based on indirect antifouling with substrate in the environment. Coating D is based on indirect antifouling with the substrates provided from the paint. Reproduced with permission.^[1] Copyright 2012, American Chemical Society. e) Schematic of UV-based antifouling. f) Simulated results, and outcomes of UV-based antifouling from open-water trials. Reproduced with permission.^[219] Copyright 2017, Clearance Center.

the mutagenic, carcinogenic, and teratogenic properties of vanadium compounds restrict their large-scale application.^[211] To address this, cerium dioxide (CeO_2) nanorods with HPO-like activity were developed as biocompatible artificial enzymes. Zhou et al. fabricated a Ce-based MOF capable of mimicking HPO activity, catalyzing the generation of hypobromous acid, and exhibiting antibacterial properties. In addition, these materials exhibit high chemical stability and low environmental toxicity, owing to the insolubility of ceria in water.^[212] In the future, other oxide nanoparticles (e.g., CeO_{2-x}) with intrinsic catalytic activity and environmental sustainability may be developed for broad applications in antifouling coatings. Although the effectiveness, safety, and toxicity of marine natural products have been investigated in laboratory and field studies, their commercialization remains challenging. Production is often costly and time-consuming, and regulatory approval for introducing new bioactive compounds

into the environment is both expensive and lengthy. Therefore, fostering collaboration between industry and research laboratories is essential to shorten the timeline from laboratory discovery to market-ready products.^[213]

Besides natural and mimic antifoulants, enzymes have been explored as non-toxic antifoulants such as proteases, carbohydrases and commercially available Alcalase.^[214] Enzymatic antifouling approaches can be categorized into direct and indirect types, depending on the nature of the enzymes and substrates involved (Figure 10d). Direct enzymatic antifouling works by degrading the fouling organism or its bioadhesive, while indirect enzymatic antifouling relies on the enzymatic generation of biocides from substrates present in seawater or coating ingredients.^[1] For example, Peres et al. developed papain-based antifouling coatings and conducted a seven-month test in the Mediterranean Sea. They discovered that papain, a

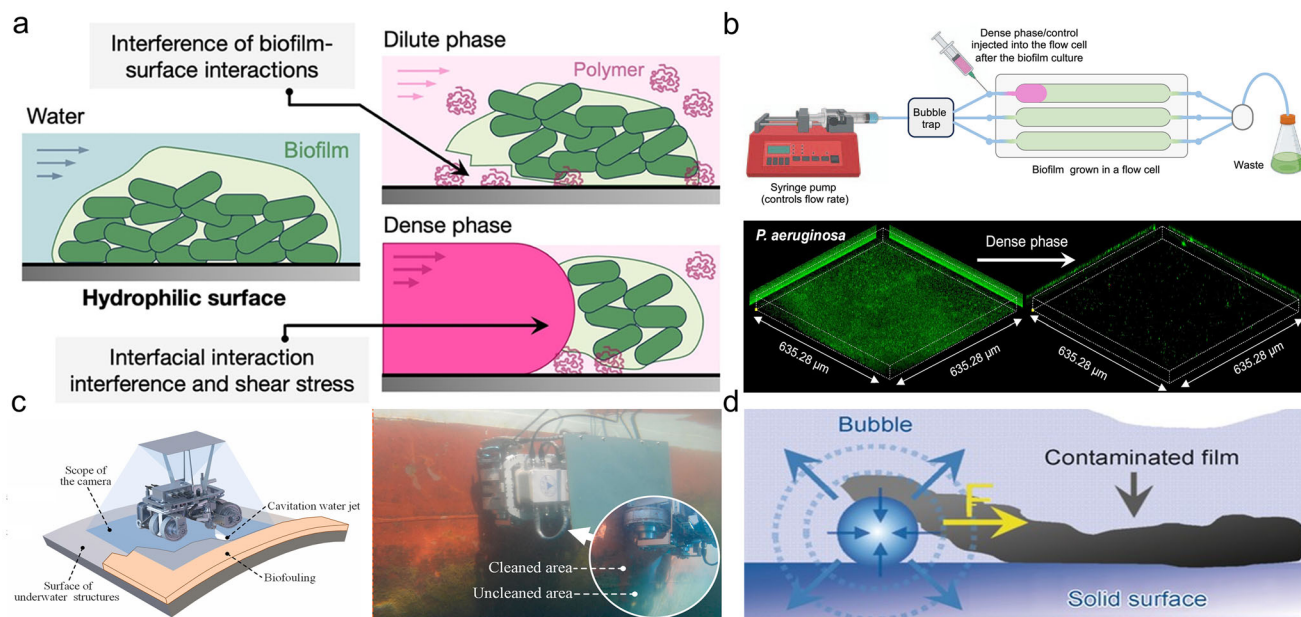


Figure 11. a) Hypothetical model for biofilm removal by the polyester based on the experimental results. Reproduced with permission.^[222] Copyright 2024, American Chemical Society. b) Experimental setup for cleaning biofilms and results of *P. aeruginosa* biofilm cleaning. Reproduced with permission.^[222] Copyright 2024, American Chemical Society. c) Schematic and field experiment of the underwater robot operation. Reproduced with permission.^[226] Copyright 2024, Elsevier. d) Schematic illustration of ultrasonic cleaning of a solid contaminated surface. Reproduced with permission.^[229] Copyright 2016, Wiley-VCH.

nature-friendly antifouling proteolytic enzyme that biodegrades easily in the marine environment, exhibited excellent antifouling properties.^[215] Kristensen et al. developed a coating composed of starch, glucoamylase, and hexose oxidase, which enables the release of hydrogen peroxide and demonstrates highly effective antifouling properties. Compared to coatings without enzyme, the enzyme-based coating significantly reduced both barnacle attachment and diatoms coverage.^[216] These findings provide a proof-of-concept for the antifouling effect of enzyme-based coatings, and show the potential to be applied in marine antifouling. However, challenges remain in developing enzyme-based coatings as ideal marine antifouling solutions. While the concepts and short-term antifouling activity of coatings have been demonstrated in several cases, long-term effectiveness against all types of fouling organisms has not yet been reported. In particular, achieving controlled release and stability of enzymes when incorporated into a coating is a significant obstacle.^[217] Thus, designing new polymeric structures that preserve enzyme activity, enhance durability, and provide broad-spectrum antifouling effects without compromising coating performance could improve their applicability and represent a key direction for future research.

To replace traditional fouling-killing strategies, emerging UV irradiation could be also used to mitigate marine biofouling by creating unfavorable microenvironment for fouling organisms (Figure 10e). Short-wavelength UVC radiation ($\approx 230\text{--}300\text{ nm}$) is lethal to microorganisms, disrupting vital cellular functions.^[218] To evaluate the antifouling effectiveness of UV lamps in real marine conditions, Salters et al. deployed identical samples across different global locations for several months. Their results showed that UV-exposed areas remained free of fouling, whereas nonilluminated samples experienced considerable bio-

fouling (Figure 10f).^[219] Ryan et al. explored the practical application of UV irradiation in marine antifouling and concluded that intermittent high-intensity UV exposure could effectively inhibit fouling.^[220] Such approaches could also be applied to subsea equipment, offshore industry structures, and transport pipelines.^[219] Future development of UV technology may focus on extending operational duration to enhance its applicability in marine antifouling. However, this could raise concerns, as prolonged UV exposure may have negative effects on materials commonly used in the marine industry.^[220] Designing polymeric coatings with durability as the substrate for UV light has the potential to not only improve antifouling durability and extend the service life but also enhance antifouling performance.

Cleaning technology is a common and classical method. Recently, emerging passive and active cleaning technologies have reintroduced themselves as promising approaches in the field of marine biofouling control. It has been observed that the barnacles secrete a phase-separating fluid rich in lipids and phenolic compounds that oxidize and detaches marine biofilms from the surface.^[221] Inspired by this, Vishwakarma et al. conceptualized that an aqueous polymer solution that outcompetes biofilm–substrate interactions in wet conditions could be a promising approach to displace existing biofilms. They developed novel biofilm-clearing materials—liquid phase-separating polyesters (coacervates)—which can spread and form strong adhesions with substrates underwater (Figure 11a). Their results demonstrated that nonionic, coacervating synthetic polymers effectively removed $\approx 99\%$ of *Pseudomonas aeruginosa* biofilm biomass from underwater surfaces (Figure 11b).^[222] This study paves the way for the design of a new class of polymers aimed at

weakening biofilm-surface interactions, thereby enhancing antifouling efficiency in marine applications.

Besides polymer cleaning methods, underwater robots can use various brushes to remove marine organisms from submerged hulls, adapting to different hull types and coatings. However, previous mechanical approaches may damage the ship's surface or coatings, compromising their integrity.^[223] To mitigate this, contactless cleaning methods have been developed, which are less damaging. For instance, laser cleaning uses rapid laser scanning of the hull to remove surface fouling effectively.^[224,225] In particular, the recently developed proactive cleaning technology activates cleaning during the biofilm stage, preventing fouling from growing.^[226–228] This represents a shift from conventional passive cleaning to active cleaning. The underwater cleaning robot integrates functions such as fouling organism identification and active removal (Figure 11c).^[226] It can promptly eliminate fouling at the early stages of localized antifouling coating failure, preventing the spread of fouling and maintaining the hull in a long-term clean state. Similarly, ultrasonic cleaning uses high-frequency ultrasound pulses to dislodge marine organisms from the surface (Figure 11d).^[229,230] The underlying mechanism is to generate microscopic agitation, which provides a gentle yet continuous effect. It keeps hulls and equipment clean by eliminating microorganisms from the surface, effectively disrupting the nutrient sources of larger organisms.^[231] Likewise, the microscopic movement of water prevents barnacle and mussel from attaching.^[232] These cleaning methods mentioned above rely on external forces to reduce fouling on the surface, which may still cause some surface damage. Therefore, designing polymeric coatings with mechanical robustness and high adhesion strength, capable of integrating such active cleaning or ultrasonic cleaning technologies, may offer a promising solution for marine antifouling.

Emerging antifouling strategies are environmentally friendly because they avoid the release of toxic substances and hold considerable potential for marine applications. However, most of these strategies have not yet been commercially implemented in real-world marine settings. Applying these emerging strategies in the marine field and evaluating their antifouling performance could be a crucial step for future development. When integrating emerging technologies with polymeric materials and interfaces, it is also essential to consider polymer structure design, as different polymer structures have varying antifouling performances and functions (Table 1). Such consideration facilitates the development of durable, sustainable, and non-toxic marine antifouling solutions under different scenarios.

6. Conclusions and Outlook

In this review, we presented recent advances in polymeric materials and interfacial designs, highlighting the relationship between their structures and properties. Chemical dynamic surfaces, composed of degradable polymers, combat marine biofouling through self-renewal mechanisms. However, such coatings are consumable, and their service life is inherently limited by their thickness. Incorporating environmentally friendly antifoulants may help extend their durability. Physical dynamic surfaces create unstable or deformable interfaces under external stimuli, effectively preventing the settlement of fouling organ-

isms. Combining dynamic surfaces with additional antifouling strategies, such as incorporating fouling-resistant groups, can enhance their performance, especially in situations where external energy is unavailable to maintain surface dynamics. FRCs often suffer from poor mechanical strength, low adhesion, and limited fouling resistance. Developing polymer–ceramic hybrid coatings that integrate fouling-resistant groups with an inorganic network can provide both enhanced mechanical strength and superior antifouling performance. Further introducing self-healing or recyclable functionalities can create sustainable coatings, representing a promising next generation of marine antifouling materials. Slippery surfaces have shown excellent fouling-release performance, yet few studies have addressed the simultaneous enhancement of mechanical strength, self-healing ability, and long-term durability. Future research should prioritize the development of slippery coatings that integrate these properties, representing a critical direction for advancing marine antifouling technologies. For hydration-based fouling-resistant coatings, achieving a balance between hydrophilicity and mechanical strength remains challenging. Integrating hydrophobic cross-linked networks with self-enrichment strategies offers a promising approach to address this issue. For other types of fouling-resistant strategies such as protein-based coatings and electrical methods, their long-term effectiveness in marine environments requires further investigation. For microstructured coatings with topographical resistance, the high cost and limited scalability hinder their widespread implementation, highlighting the need for simple and cost-effective methods to generate such structures. In addition to traditional antifouling coatings, emerging strategies have been investigated, including natural antifoulants, enzymatic antifouling, ultraviolet (UV) radiation, and passive and active cleaning technology. These emerging methods aim to achieve environmentally friendly, cost-effective, and scalable solutions for marine antifouling applications. Designing optimal polymeric structures in conjunction with these emerging methods will not only improve their antifouling efficiency but also extend their service life and enhance overall performance.

Despite substantial advances in marine antifouling coatings and strategies, several challenges continue to limit their practical application. The marine environment is harsh and complex, with factors such as temperature, salinity, and pressure affecting coating performance. In addition, the vast diversity of biofouling organisms requires coatings capable of preventing the attachment of multiple species.^[233] Long-term durability is also essential because marine infrastructures must maintain functionality over extended periods.^[76] Sustainability poses an additional challenge because conventional antifouling coatings often contain toxic substances that can adversely affect the environment and marine ecosystems.

The antifouling coatings and strategies reviewed here—including those that release, resist, or kill biofouling—play a crucial role in marine antifouling. However, relying on a single antifouling strategy often results in limited effectiveness in the complex marine environment. Therefore, there is an urgent need to develop synergistic coatings that integrate multiple antifouling mechanisms, providing enhanced and long-lasting performance. Specifically, synergistic combinations of fouling release and fouling resistance may show the most immediate potential to be used as durable, sustainable and non-toxic marine

Table 1. Summary of antifouling coatings and strategies.

Materials	Antifouling strategies	Performance	Refs.
Poly(ethylene adipate) (PEA) based polyurethane with DCOIT	Chemical dynamic surface and antifoulant	Fouling rating after immersion 6 months in the seawater over 90	[44]
Copolymer of MDO, DEM, and KH570	Chemical dynamic surface and fouling resistance	Reduce the fibrinogen adsorption over 97%	[48]
Copolymer of MDO and MMA with DCOIT	Chemical dynamic surface and antifoulant	Fouling rating after immersion 3 months in the seawater over 90	[47]
Copolymer of MDO, MMA, and TCBSA	Chemical dynamic surface and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 8\%$	[50]
Copolymer of MDO, MMA, and BIT	Chemical dynamic surface and antifoulant	Fouling rating after immersion 6 months in the seawater over 90	[49]
Hyperbranched copolymers of MMA, PCL- V_2 , and TCB	Chemical dynamic surface and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 0\%$	[57]
Poly(lactic acid)-based polyurethane with butenolide	Chemical dynamic surface and antifoulant	After immersion in South China Sea for 3 months, the area covered by biofouling $\approx 7\%$	[207]
Hyperbranched copolymers of MDO, VAc, and diethylene glycol divinyl ether	Chemical dynamic surface and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 0\%$	[151]
Hyperbranched copolymers of methacrylic anhydride and TCB-TCPM	Chemical dynamic surface, fouling resistance, and antifoulant	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $< 1\%$	[59]
Silicone elastomer actuated by electrical stimuli	Physical dynamic surface	Detachment of <i>Cobetia marina</i> biofilms $> 80\%$	[63]
PDMS, silicone oil, and ferromagnetic carbonyl iron microparticles	Physical dynamic surface	The areas of <i>E. coli</i> coverage 91.2% lower than that on the static surface	[64]
PDMS sweepers	Physical dynamic surface	Removal rate could reach up to 99.5%	[70]
PDMS based polyurea with perfluoroalkyl modified nanodiamond	Fouling release	After immersion in natural seawater for four months, only 3% of the area was covered after cleaning	[83]
PDMS based polyurethane with TCPM pendant groups	Fouling release and antifoulant	About 5% barnacle cyprids settlement	[86]
PDMS with telomer of DFMA, triclosan acrylate, and KH590	Fouling release and antifoulant	Diatom density of ≈ 50 cell mm^{-2}	[84]
PDMS with telomer of DFMA, PEGMA, and KH590	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 0\%$	[91]
Polysiloxane crosslinked by triethoxyoctylsilane, 3-mercaptopropyl triethoxysilane, and telomer of DFMA and PEGMA	Fouling release and fouling resistance	Adhesion density of diatom < 80 cells mm^{-2}	[117]
Polysiloxane crosslinked by amine-terminated hyperbranched polysiloxane and epoxy-zirconium particle	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 6.5\%$	[120]
PDMS cross-linked via coordination bonds between BET and zinc ions	Fouling release and antifoulant	Diatom density of ≈ 67 cell mm^{-2}	[99]
Cross-linked PDMS and TCBF	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 5\%$	[96]
PDMS with UPy groups and telomer of DFMA and PEGMA	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $\approx 6\%$	[100]
Polysiloxane crosslinked by epoxy-oligosiloxane nanoclusters with amphiphilic telomer and amine-terminated curing agents	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $< 10\%$	[121]
Polysiloxane crosslinked by POSS, isocyanate-terminated polyurethane, and amphiphilic polymer	Fouling release and fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adsorption $< 10\%$	[116]
PFOTS and GPTMS grafted surface	Fouling release	$\approx 78\%$ reduction in fibrinogen adsorption in comparison to uncoated surfaces	[137]
Zwitterionic polycarboxybetaine hydrogel	Fouling resistance	The relative bacteria <i>P. aeruginosa</i> adhesion $\approx 3.2\%$	[155]
Zwitterionic triple-network hydrogel	Fouling resistance	Decreasing 99.5% of fibrinogen adsorption	[154]
PEG grafted surface	Fouling resistance	Fibrinogen adsorption on surface coverage $\approx 0.2\%$	[145]
Drug-loaded hydrogel-anchored MOF-based coating	Fouling resistance and antifoulant	Antibacterial rate of 100% <i>P. aeruginosa</i>	[161]

(Continued)

Table 1. (Continued)

Materials	Antifouling strategies	Performance	Refs.
TCBSi hydrogel paints	Fouling resistance	The relative density of diatoms on the coating surface ≈5 %	[163]
Fluorine-free amphiphilic copolymer	Fouling resistance	The coating reduces <i>Pseudomonas aeruginosa</i> attachment by over 90% compared to aluminum surfaces	[88]
Zwitterionic polysilazane coatings	Fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adhesion < 10%	[95]
Self-regenerating zwitterionic hydrogel coating	Fouling resistance	The relative marine bacteria <i>Pseudomonas</i> sp. adhesion ≈0%	[156]

antifouling coatings. Furthermore, integrating emerging technologies such as active cleaning, ultrasonic cleaning or UV irradiation technologies—which impose high requirements on mechanical robustness, adhesion strength, and durability of the surface—with polymer–ceramic hybrid coatings can enable the development of sustainable, eco-friendly marine antifouling solutions while maintaining excellent antifouling performance. It should be noted that a considerable gap exists between laboratory testing and real-world marine validation. Coatings that perform well under controlled laboratory conditions may fail to maintain the same level of effectiveness when exposed to actual marine environments. With the advancement of characterization techniques, it is increasingly important to integrate these methods into the evaluation of marine antifouling coatings, helping to bridge the gap between laboratory studies and real-world applications. Marine antifouling represents a complex systems challenge that requires knowledge across multiple disciplines. As interdisciplinary collaboration becomes ever more critical for scientific progress, incorporating computer simulations and artificial intelligence into material development offers a promising pathway for the efficient design of innovative marine antifouling coatings.

In summary, although numerous antifouling coatings and strategies have been developed, considerable opportunities remain to explore durable, sustainable, and non-toxic polymeric materials and interfaces with superior antifouling performance for marine applications. This review aims to provide readers with a comprehensive overview of recent advances in marine antifouling polymeric materials and interfaces, and to inspire the development of innovative fabrication techniques and new materials that can support both fundamental research and practical applications in this field.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

dynamic surface, fouling-release, fouling-resistance, marine antifouling, polymeric materials and interfaces

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